SHETH L.U.J. AND SIR M.V. COLLEGE

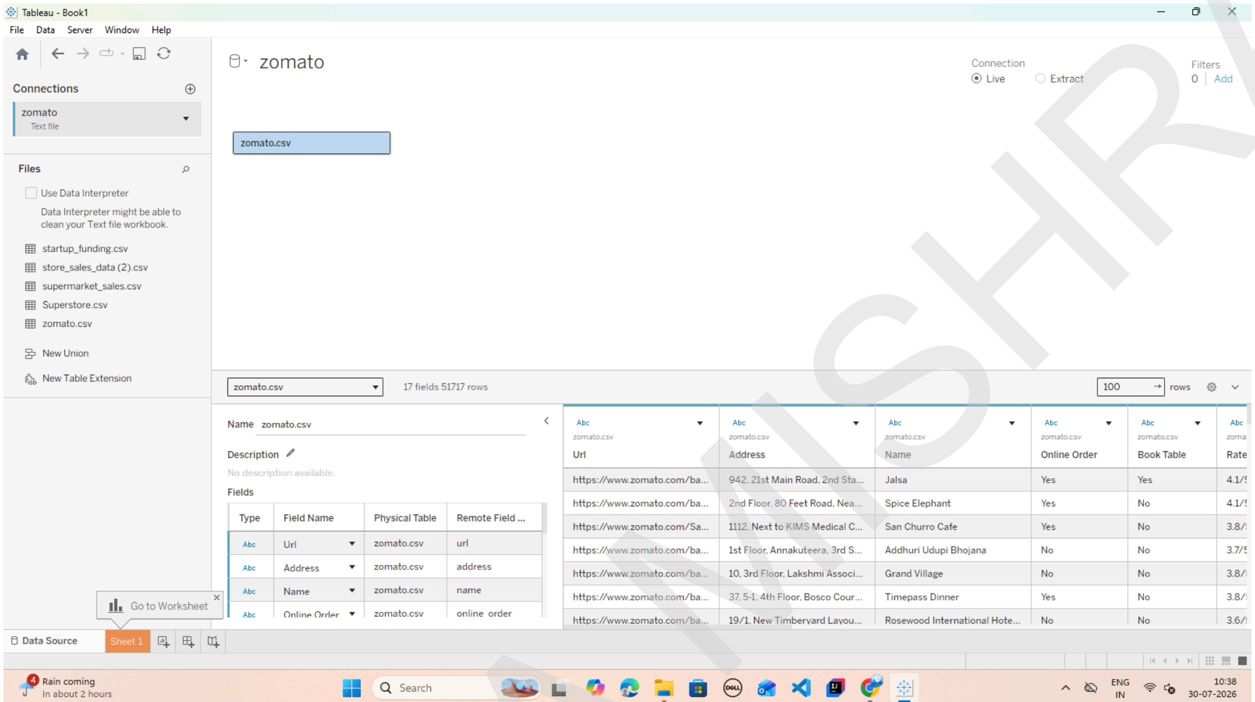
SUBJECT: Data Visualization with Power BI and Tableau

PRACTICAL NO. 5

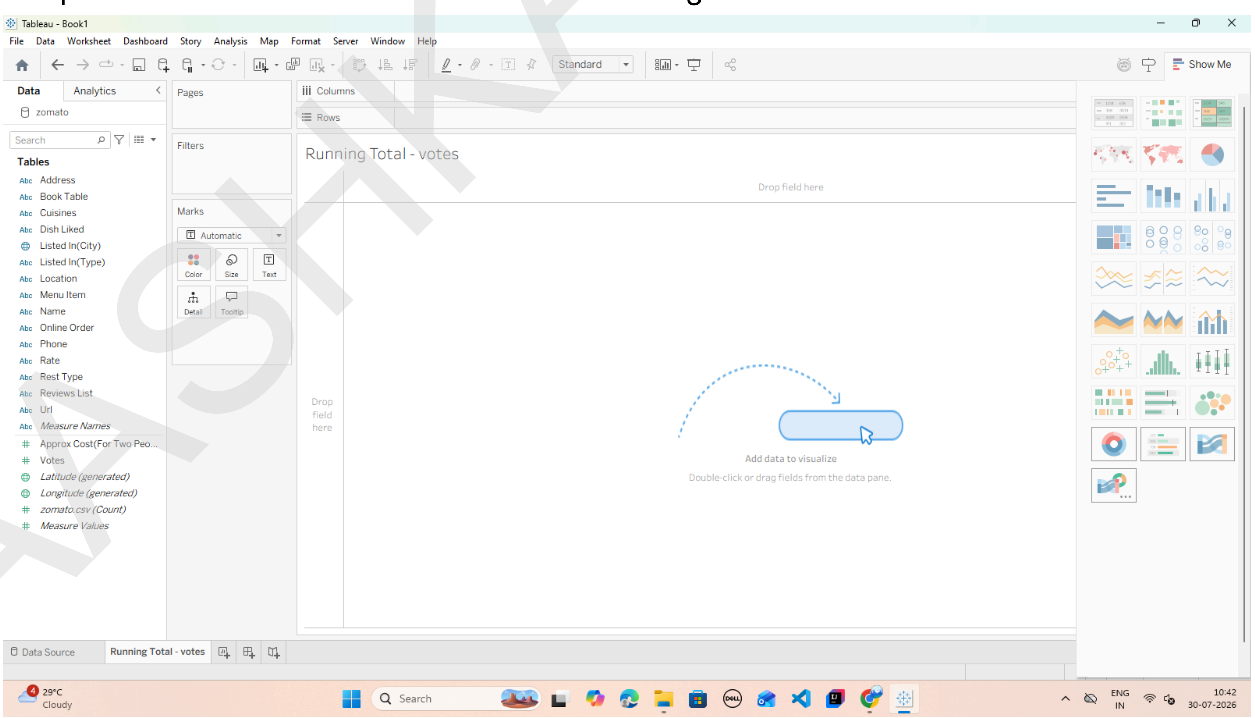
AIM: ANALYZING DATA USING TABLE CALCULATIONS

A: Apply quick table calculations (running total, percent of total)

Task 1: Create a Running Total of Votes by City (A Running Total calculates a cumulative sum across a dimension (e.g., adding each city's votes to the running sum of all previous cities)).



1. Open a new worksheet and rename it: Running Total - Votes.



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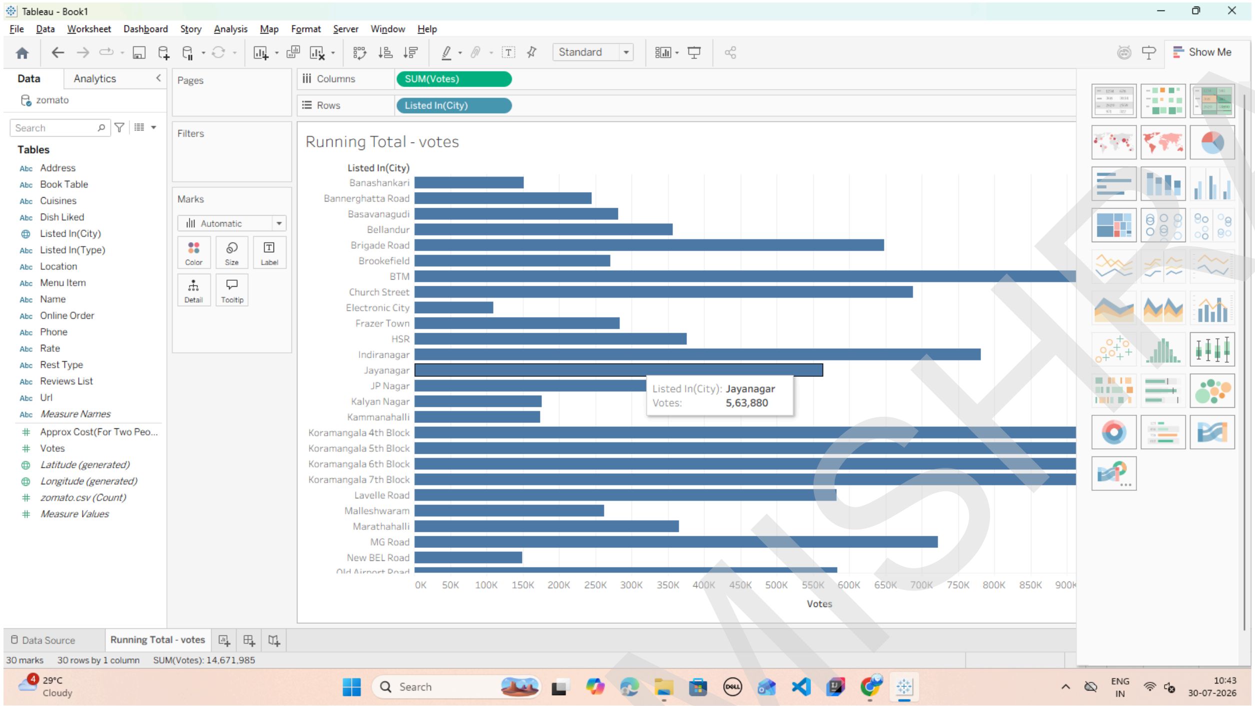
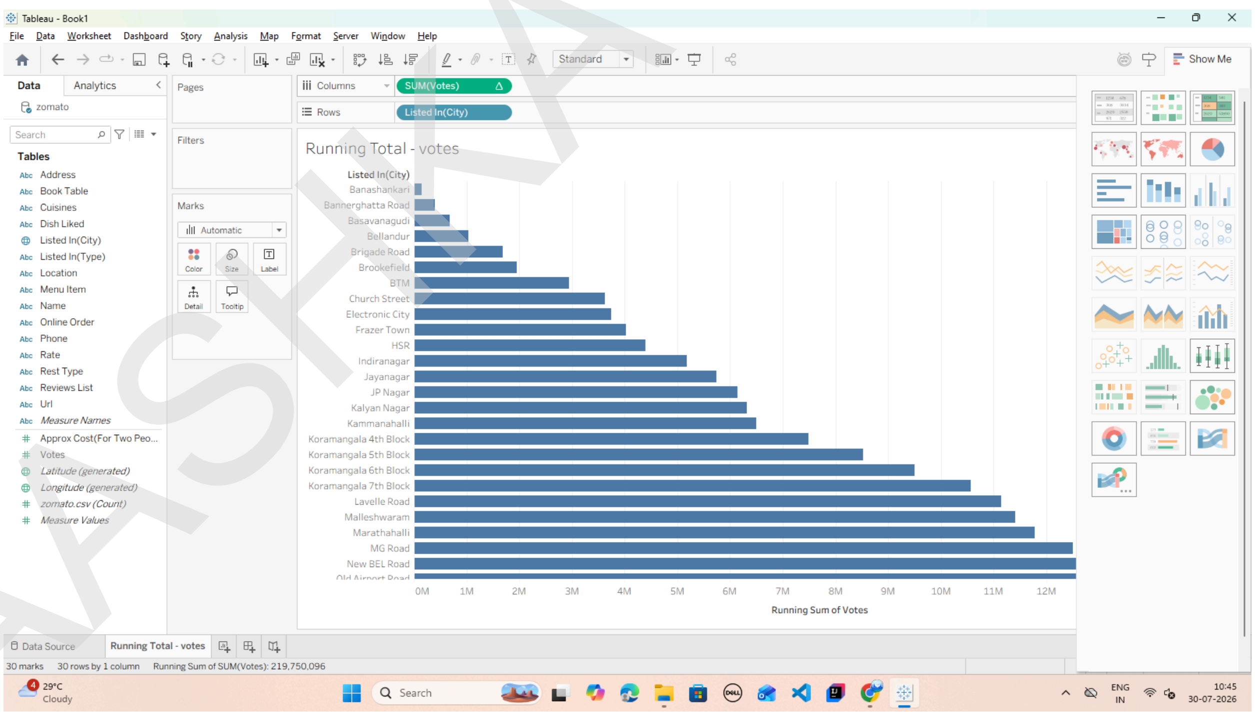
2. From the Data Pane, drag listed_in(city) to the Rows shelf.
3. Drag votes to the Columns shelf.



4. Convert to Quick Table Calculation:



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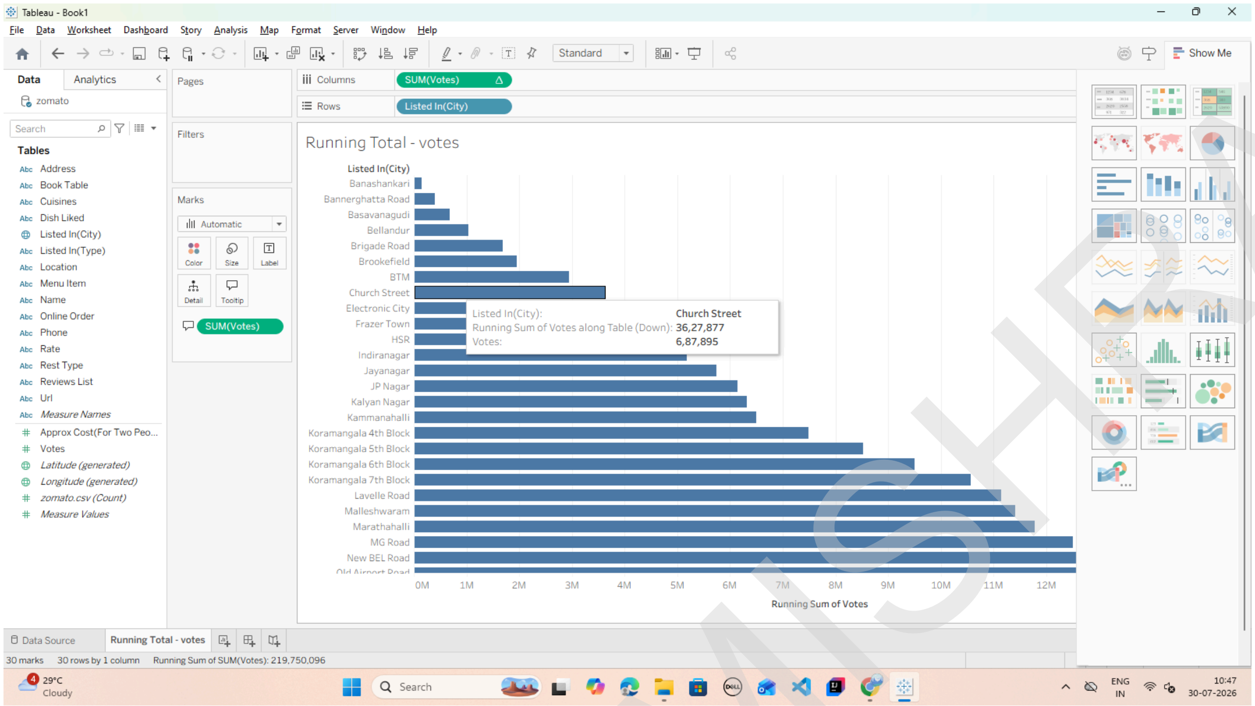
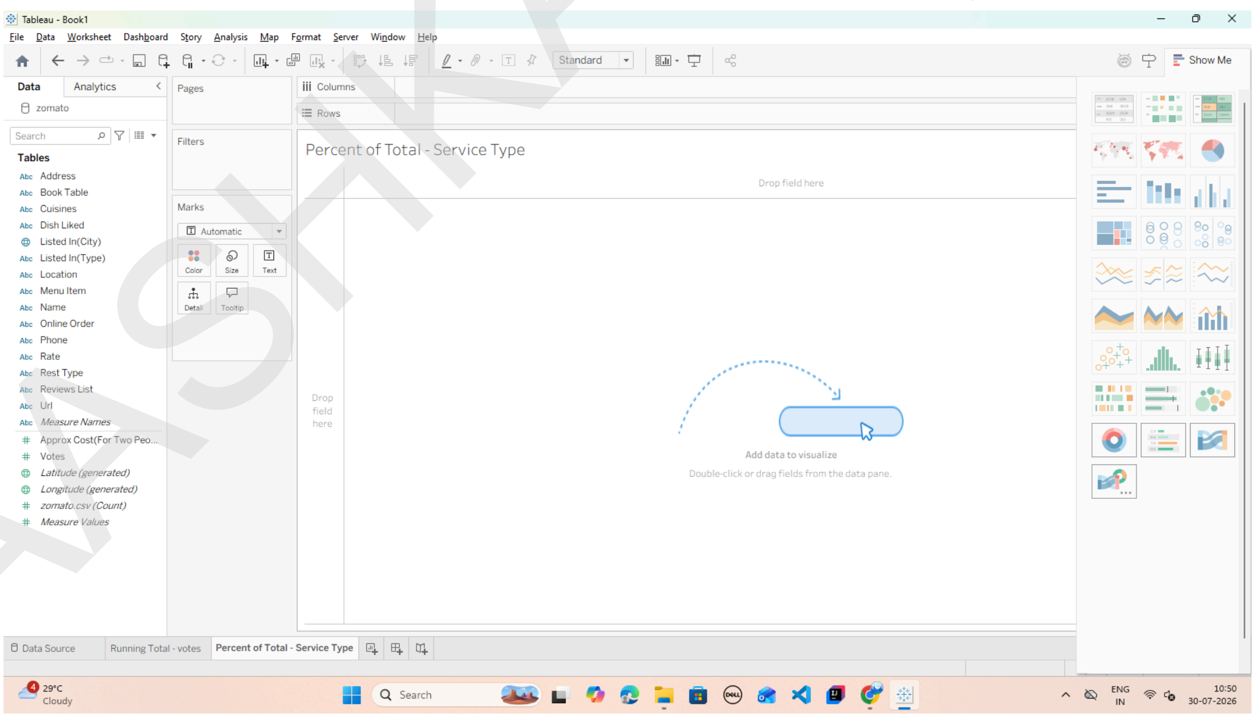
5. Add Visual Context (Secondary Measure):



Task 2: Create a Percent of Total across Categories

A Percent of Total transforms raw numbers into relative percentages of the entire view (summing to 100%).

1. Open a new worksheet and rename it: Percent of Total - Service Type.



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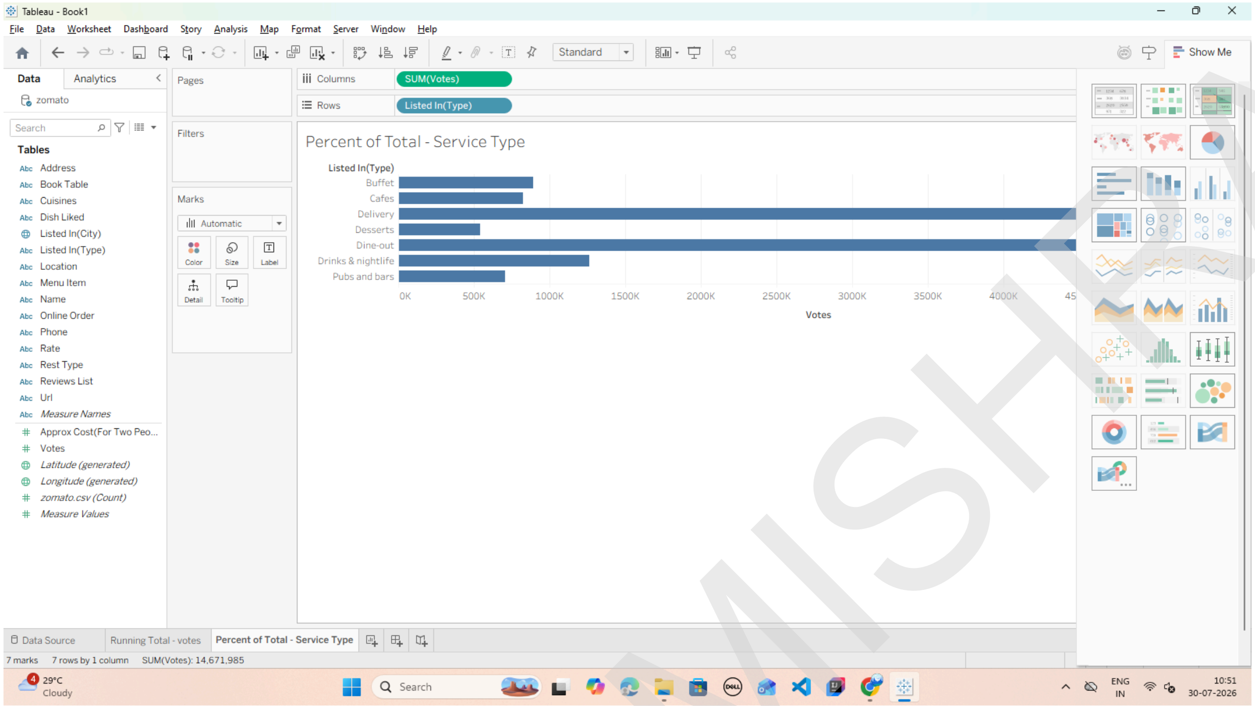
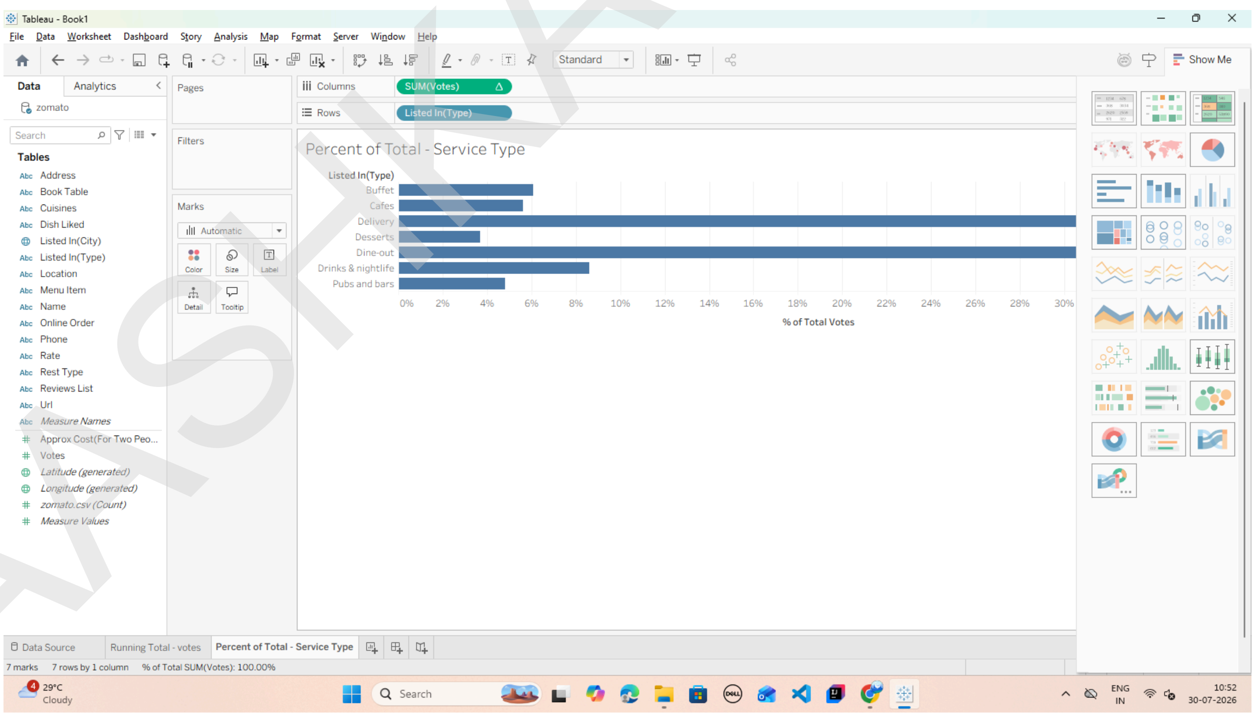
2. Drag listed_in(type) (e.g., Buffet, Cafes, Delivery, Dining) to the Rows shelf.
3. Drag votes to the Columns shelf.



4. Right-click the SUM(votes) pill on the Columns shelf.
5. Hover over Quick Table Calculation and select Percent of Total.



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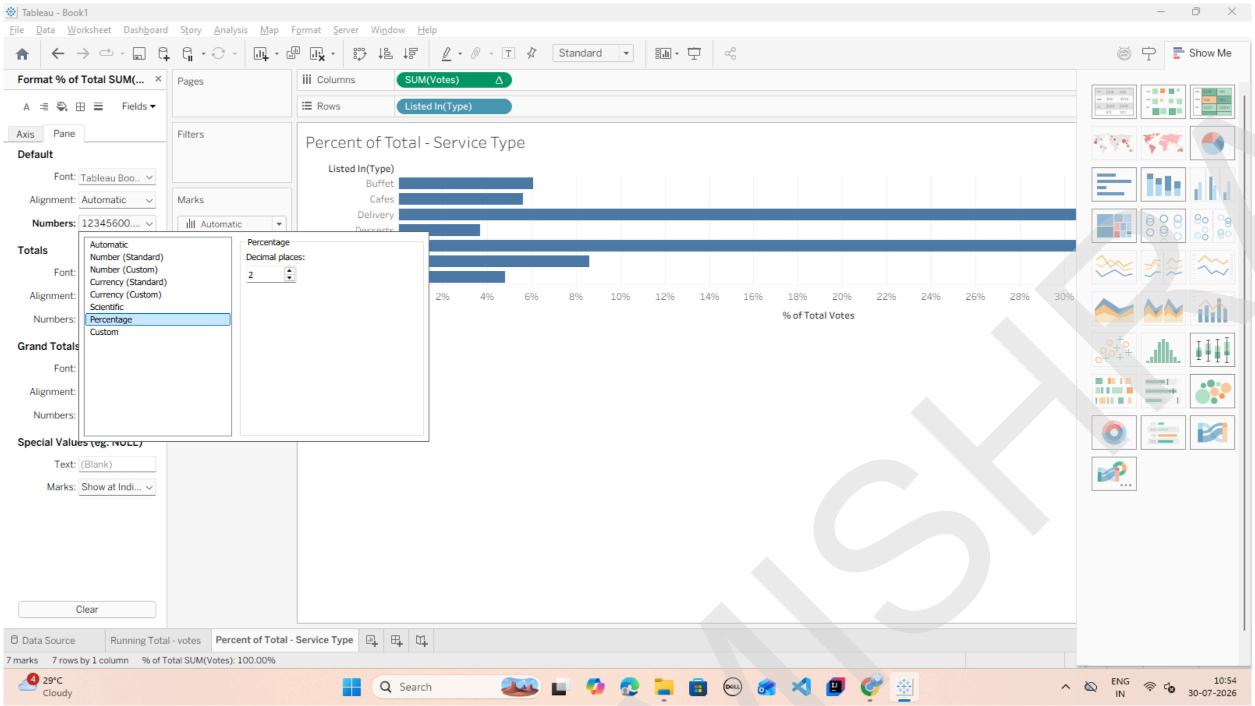
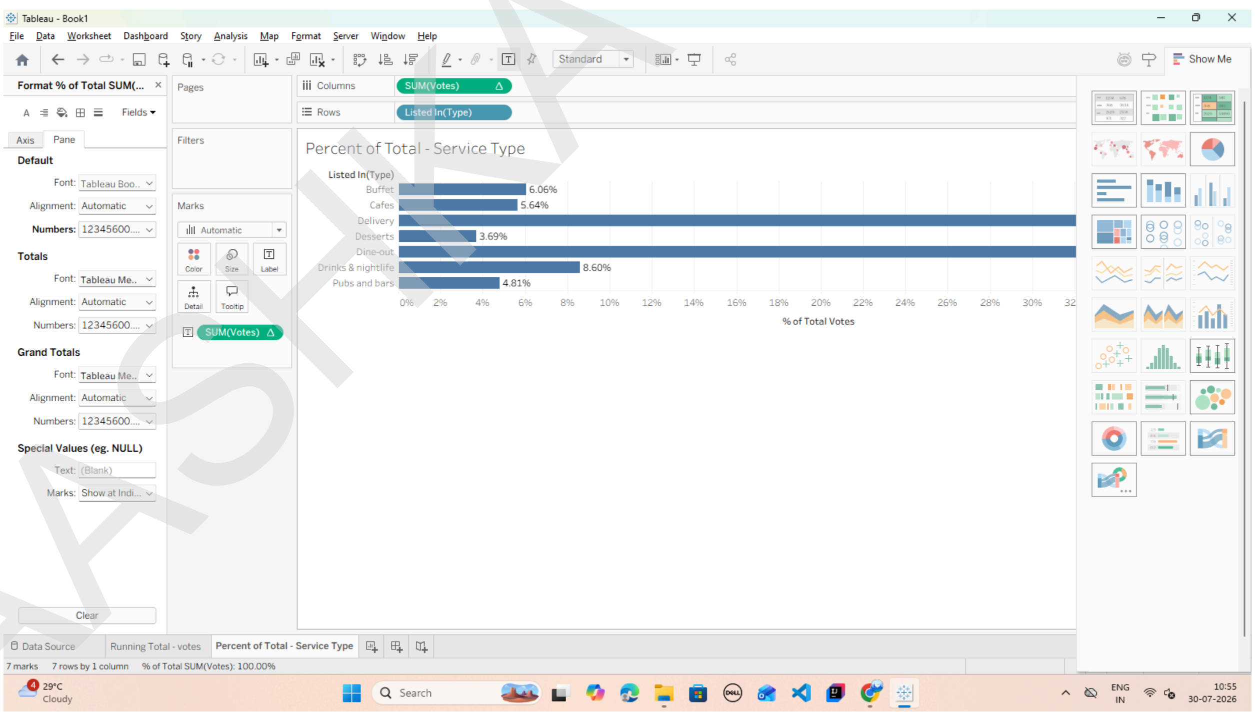
6. Format as Percentage:



7. Drag SUM(votes) onto the Label mark so the exact percentages display on top of each bar.



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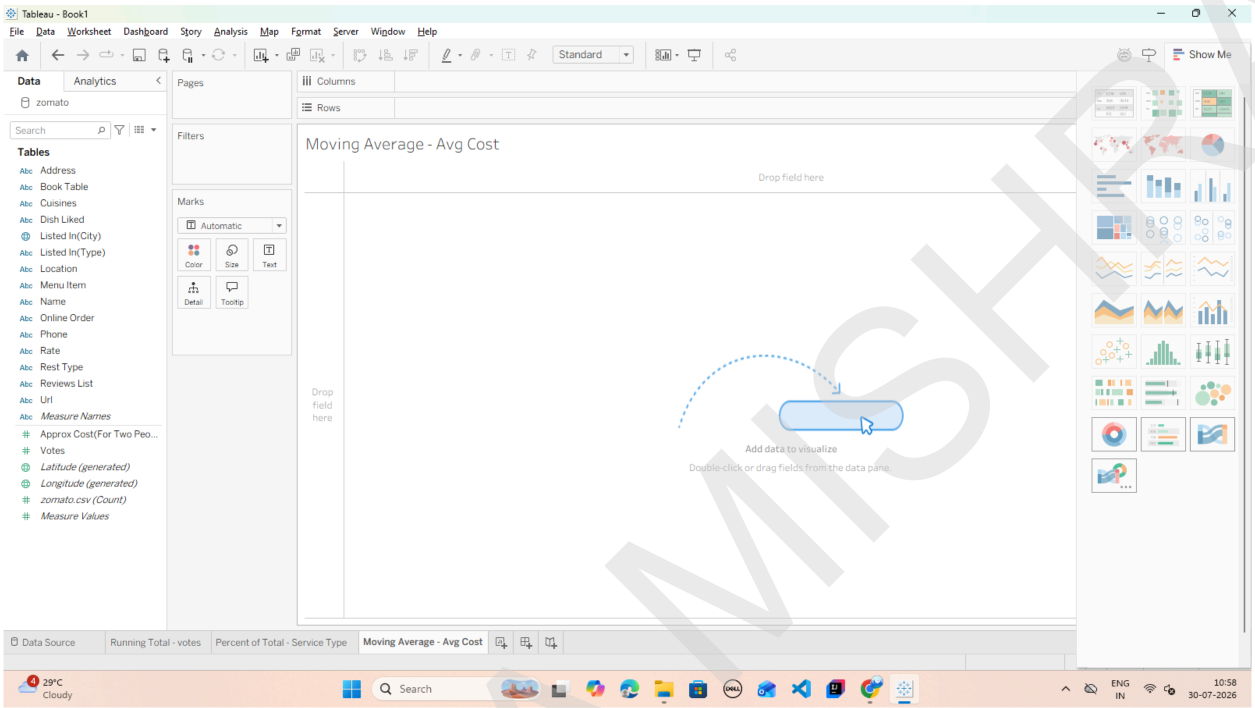
B: Create moving averages and cumulative metrics

Task 1: Create a Moving Average Calculation

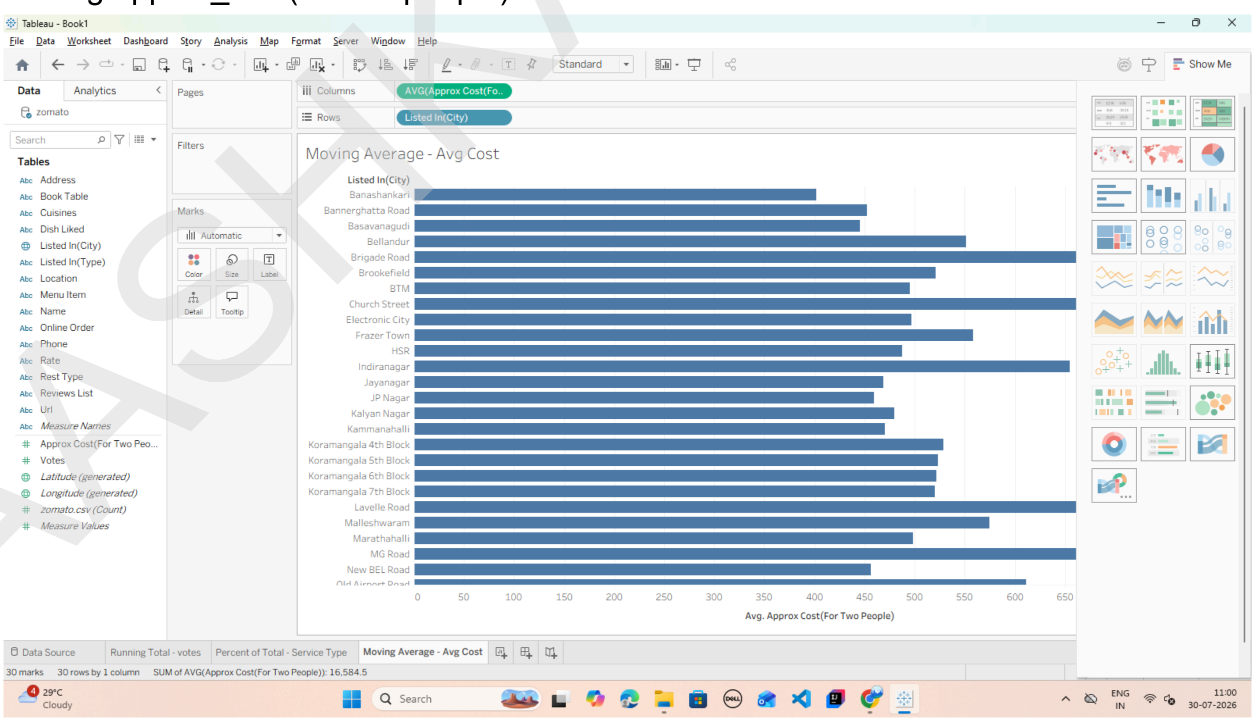
We will create a moving average to smooth out variations in pricing across restaurants based on vote counts or listing order.

1. Open a new worksheet and rename it: Moving Average - Avg Cost.



2. Drag listed_in(city) to the Rows shelf.

3. Drag approx_cost(for two people) to the Columns shelf.



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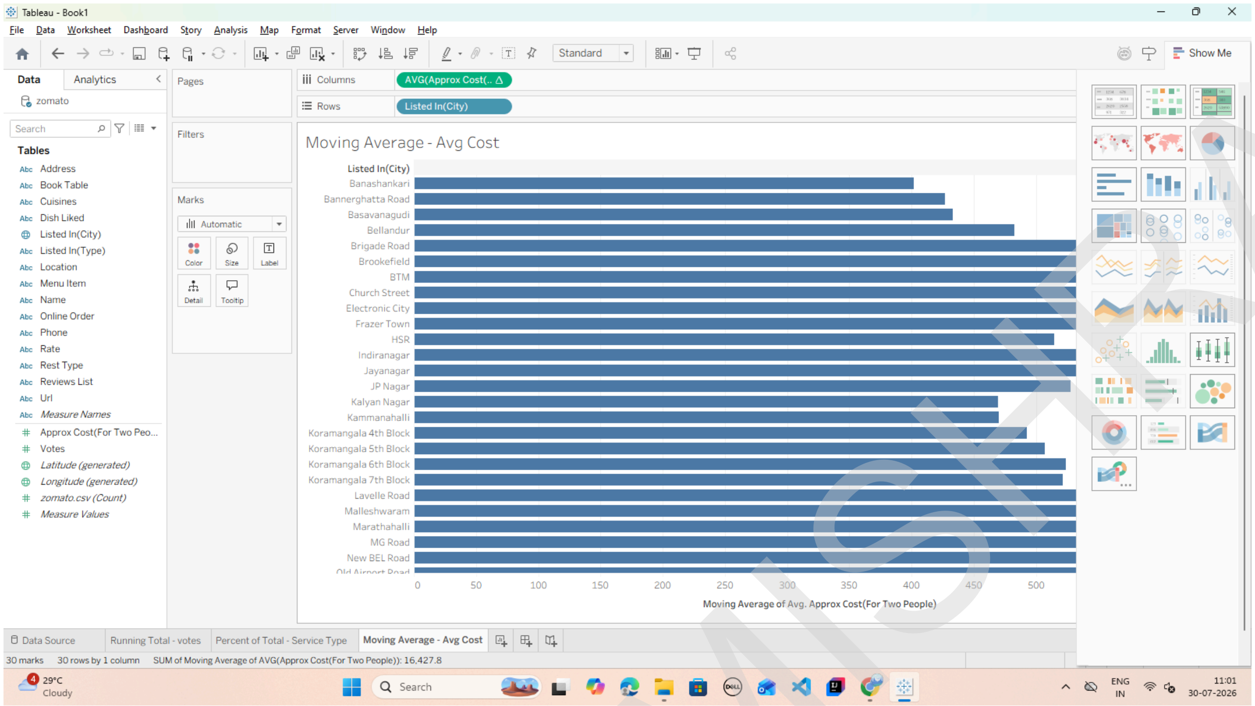
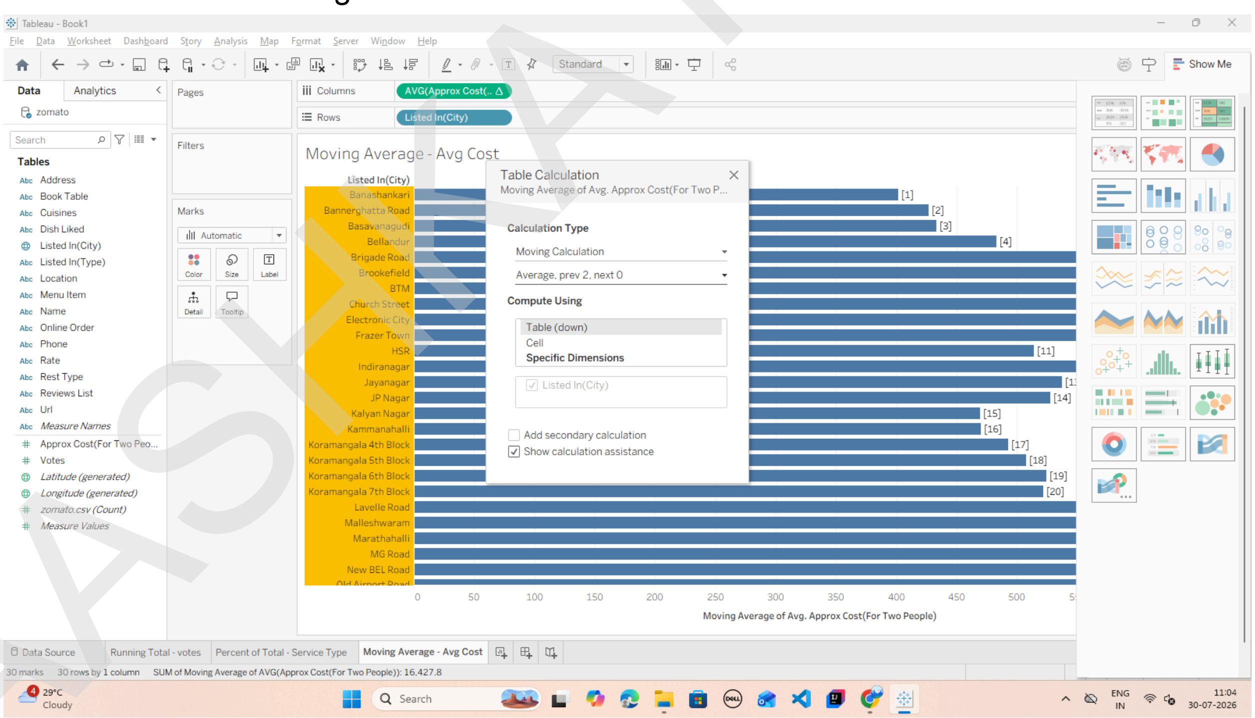
4. Convert to a Moving Average Quick Table Calculation:



5. Customize the Moving Window:



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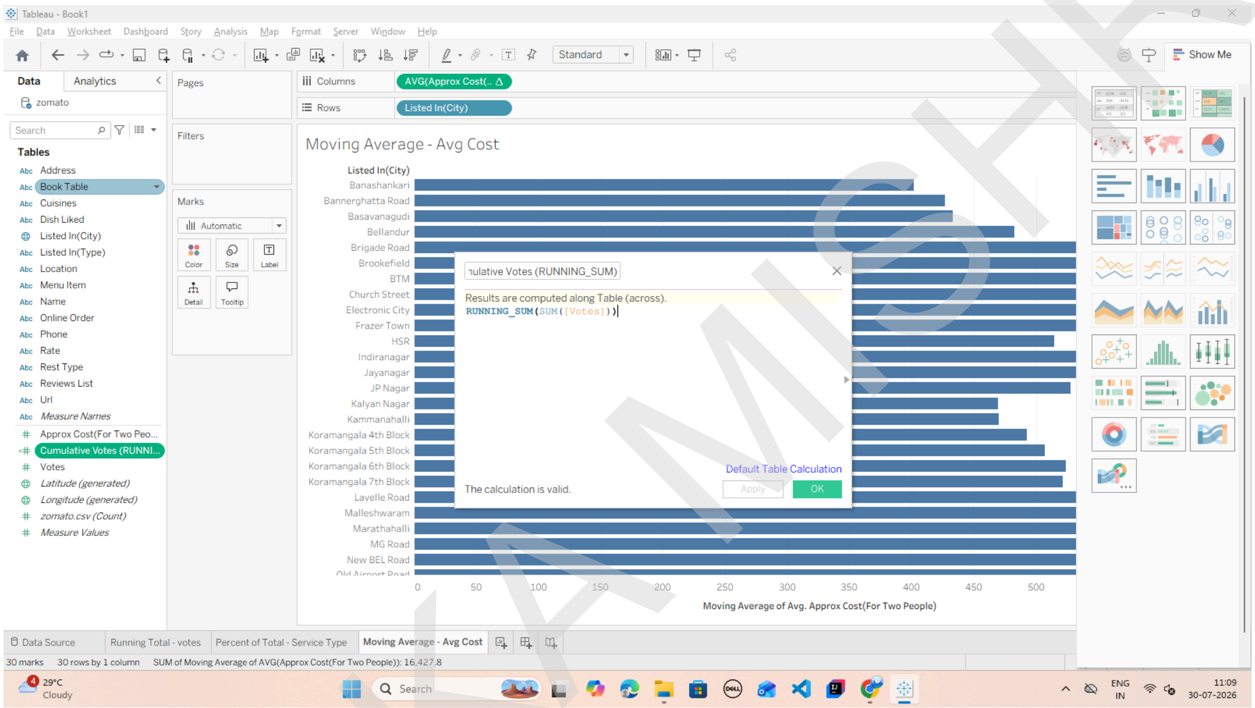
Task 2: Create a Custom Cumulative Metric via Calculated Field

Instead of using Quick Table Calculations, we can write a dedicated Table Calculation function formula directly.

1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field.
2. Name the field: Cumulative Votes (RUNNING_SUM).
3. Enter the following Table Calculation formula: RUNNING_SUM(SUM([votes]))



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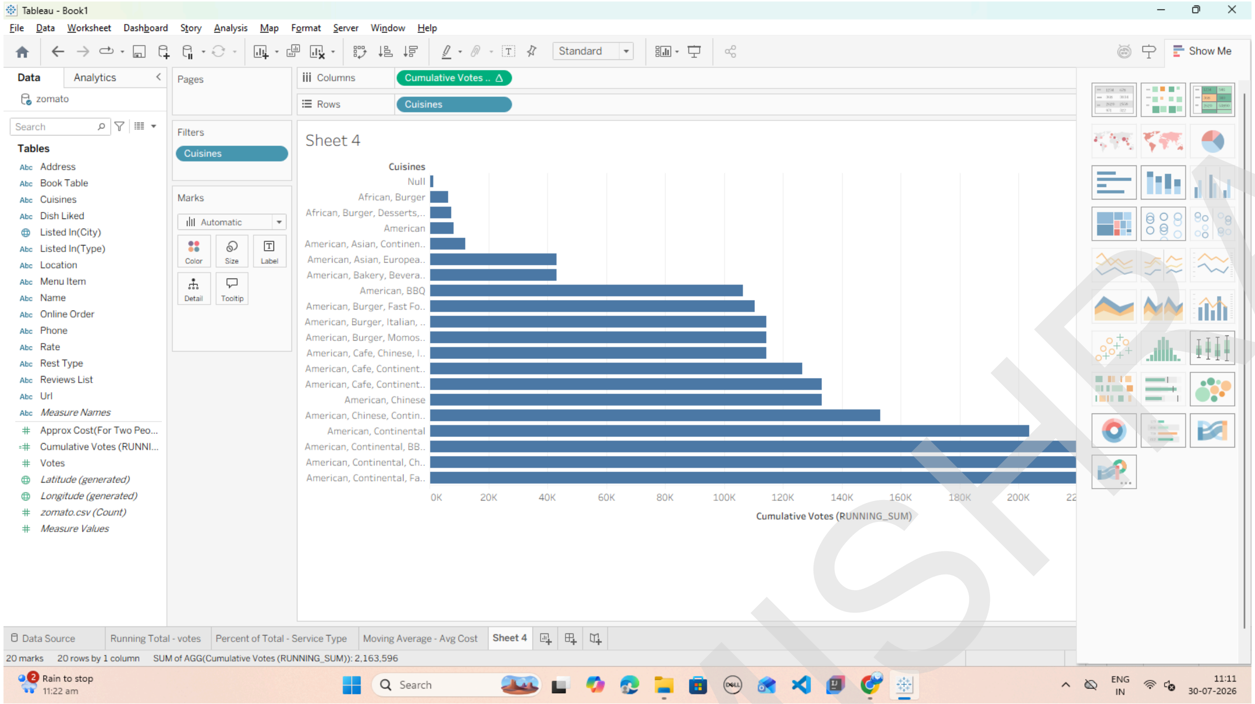
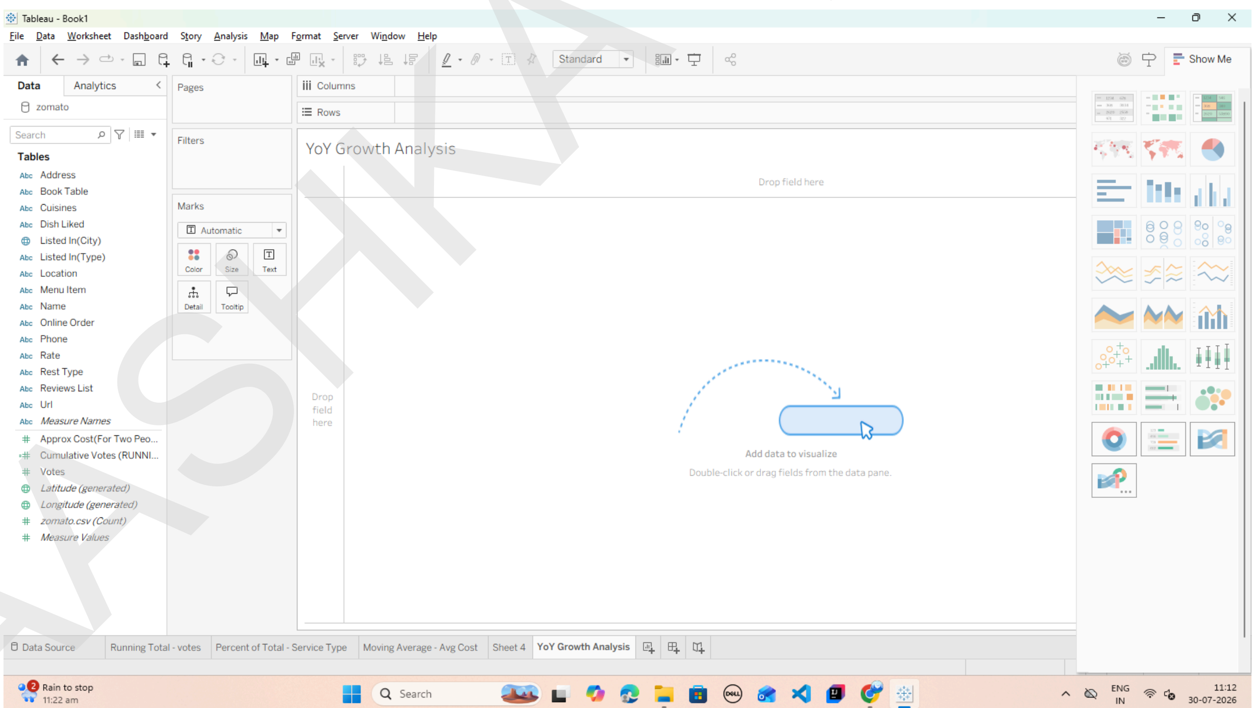


C: Perform year-over-year growth analysis

Task: Calculate Year-over-Year (YoY) / Period-over-Period Percent Growth

1. Open a new worksheet and rename it: YoY Growth Analysis.



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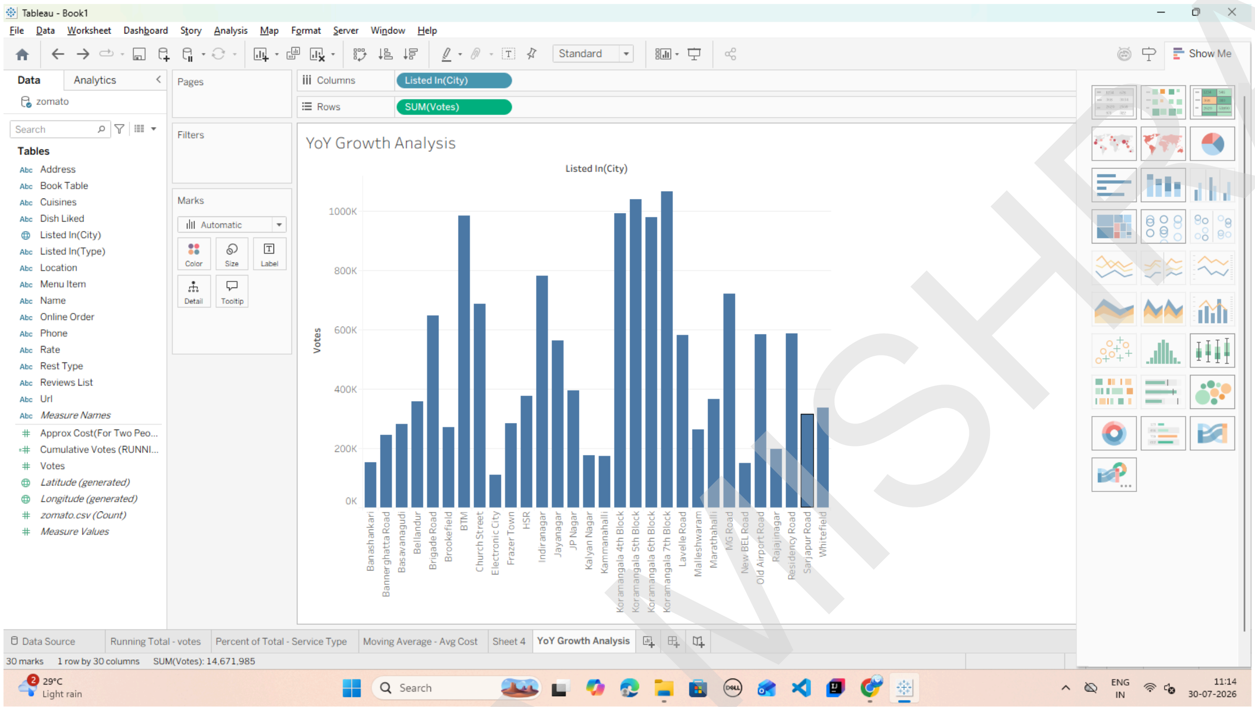
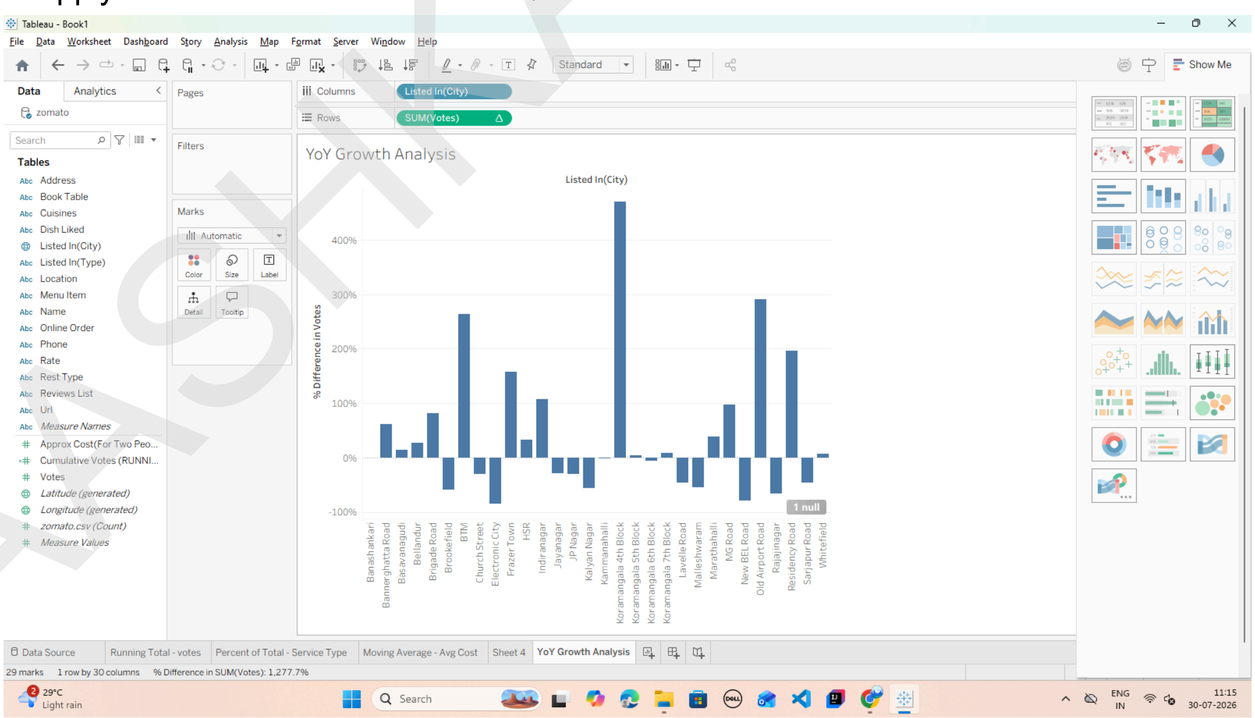
2. Place your time dimension (or ordinal grouped column like listed_in(city)) on the Columns shelf.

3. Drag votes (or approx_cost(for two people)) onto the Rows shelf as SUM(votes).



4. Apply the Year-over-Year Growth Quick Table Calculation:



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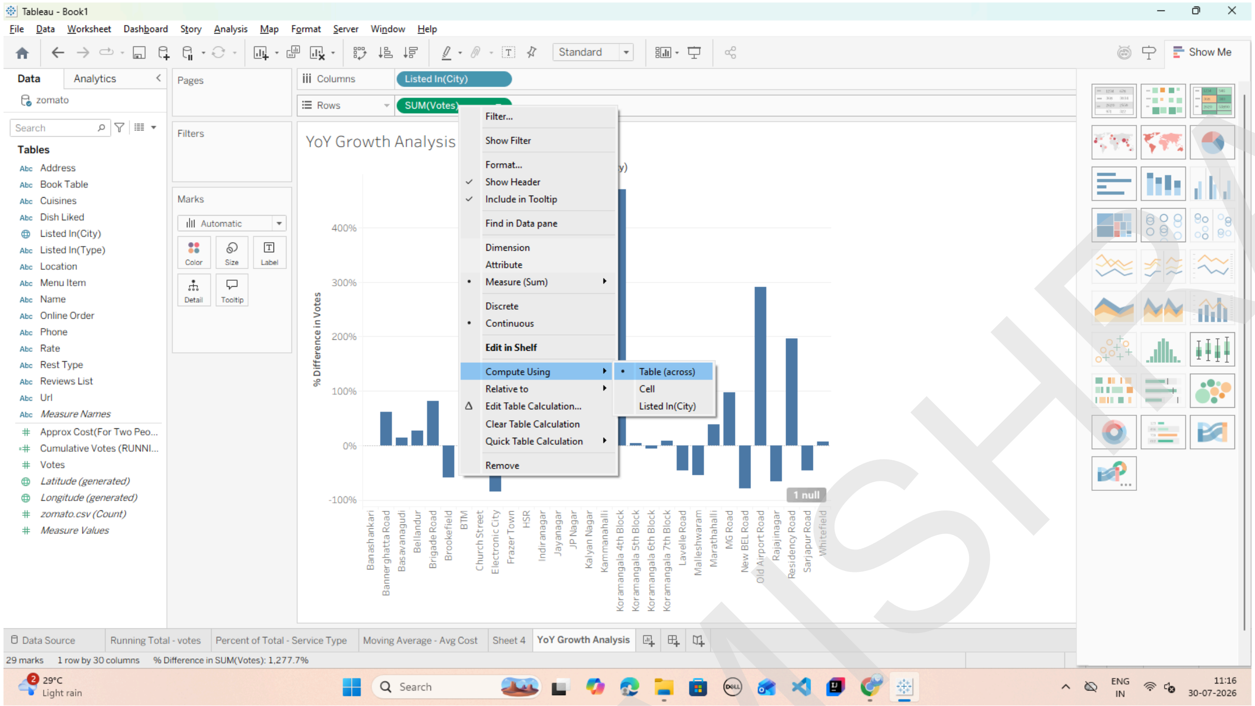
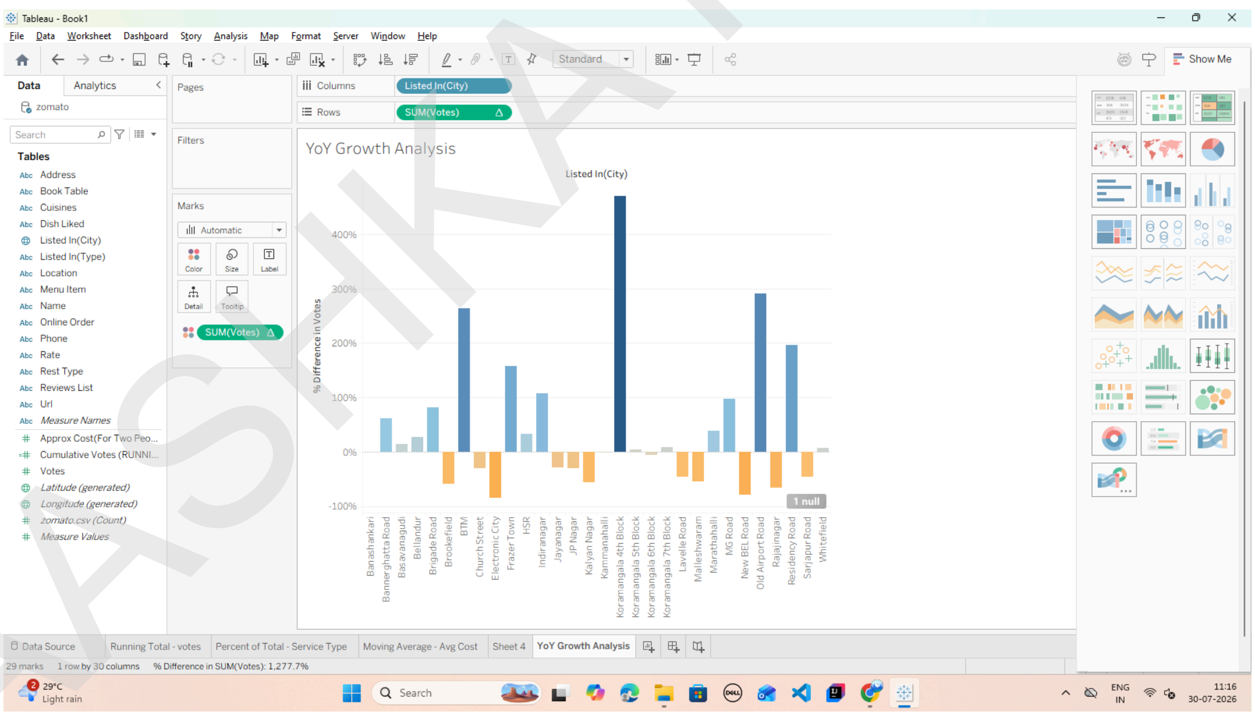
5. Configure the Growth Calculation Direction:



6. Add Visual Color Indicators:



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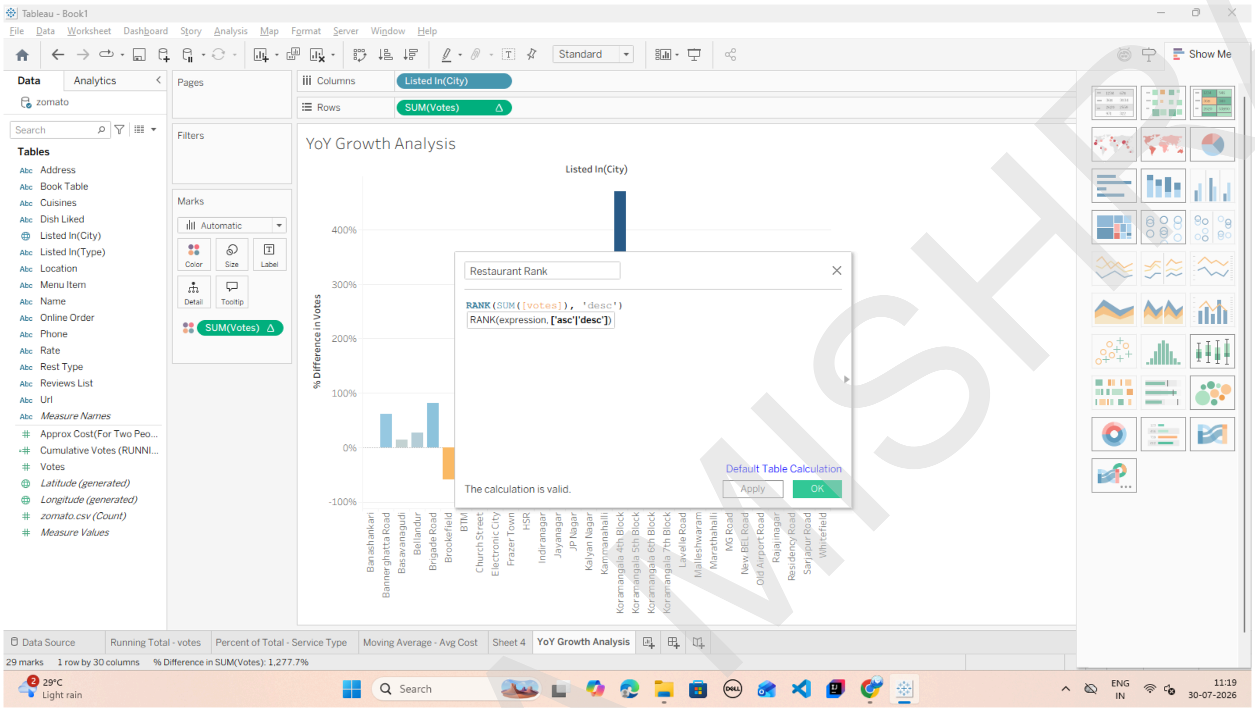
D: Use ranking functions (Top N, Bottom N) Ranking table calculations evaluate each row's aggregate value relative to all other rows in the partition, assigning a numeric rank (1, 2, 3...).

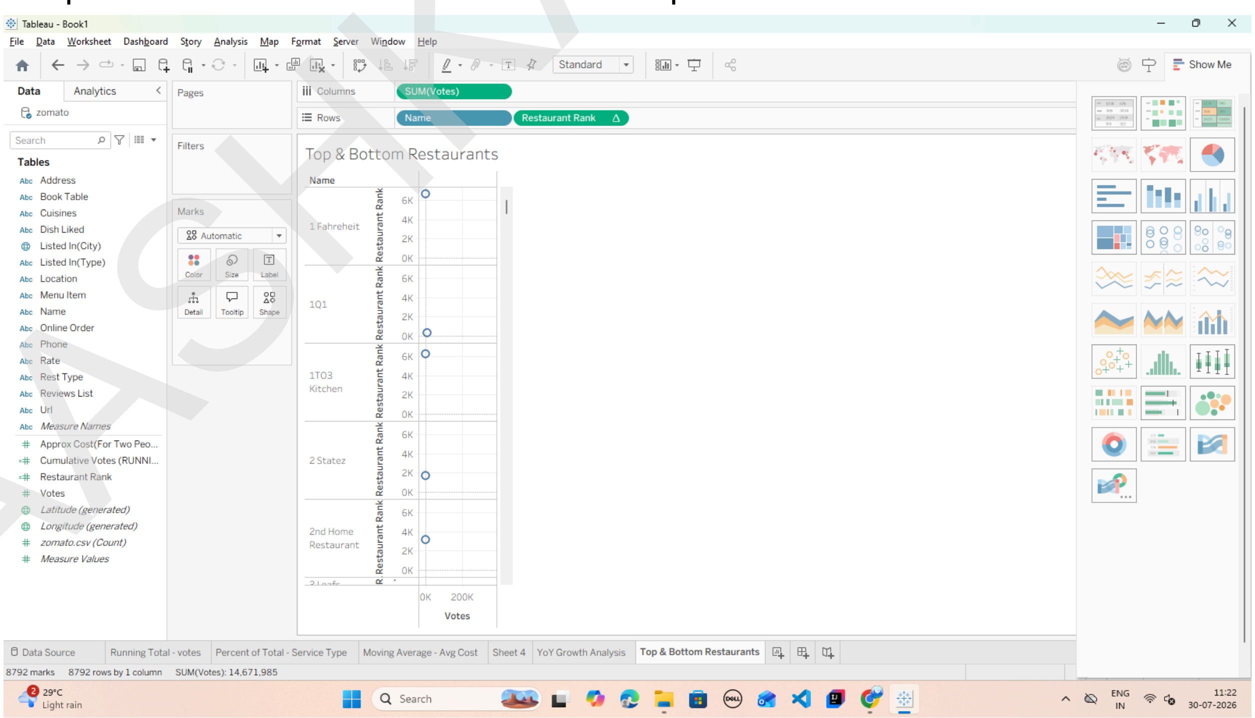
Task 1: Create a Rank Calculation

1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field.



Task 2: Build the Top N / Bottom N Ranked View

1. Open a new worksheet and rename it: Top & Bottom Restaurants.



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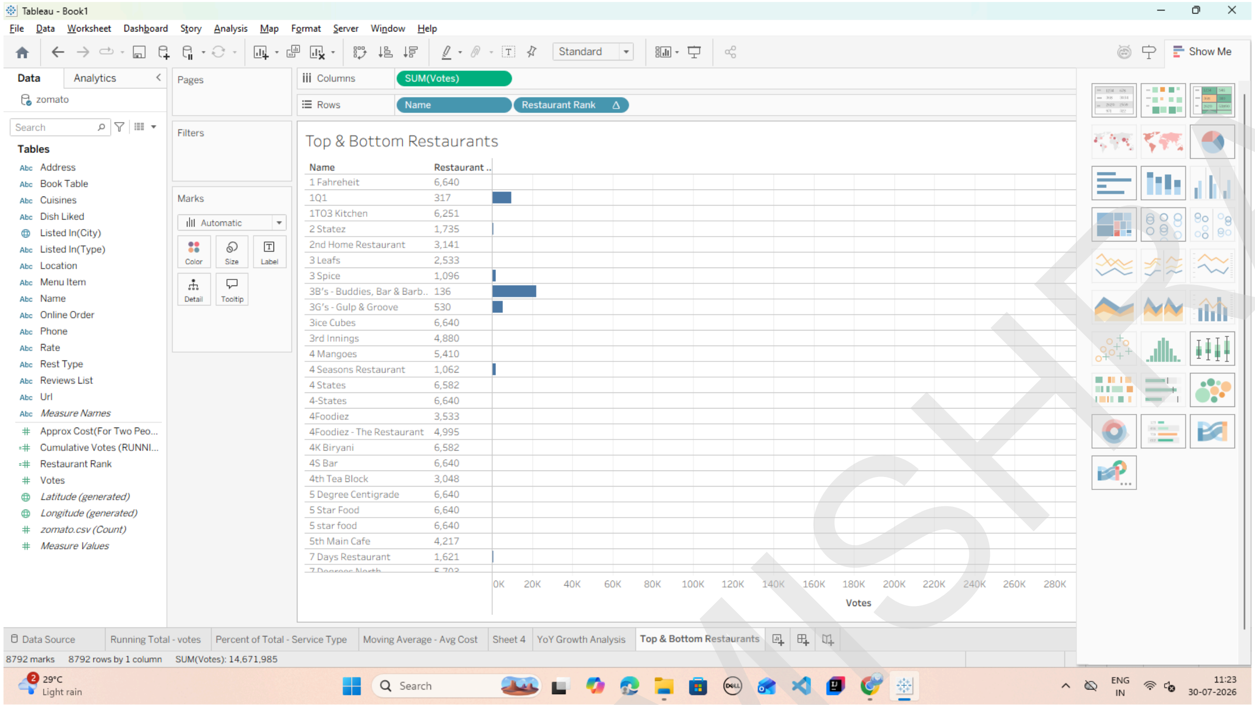
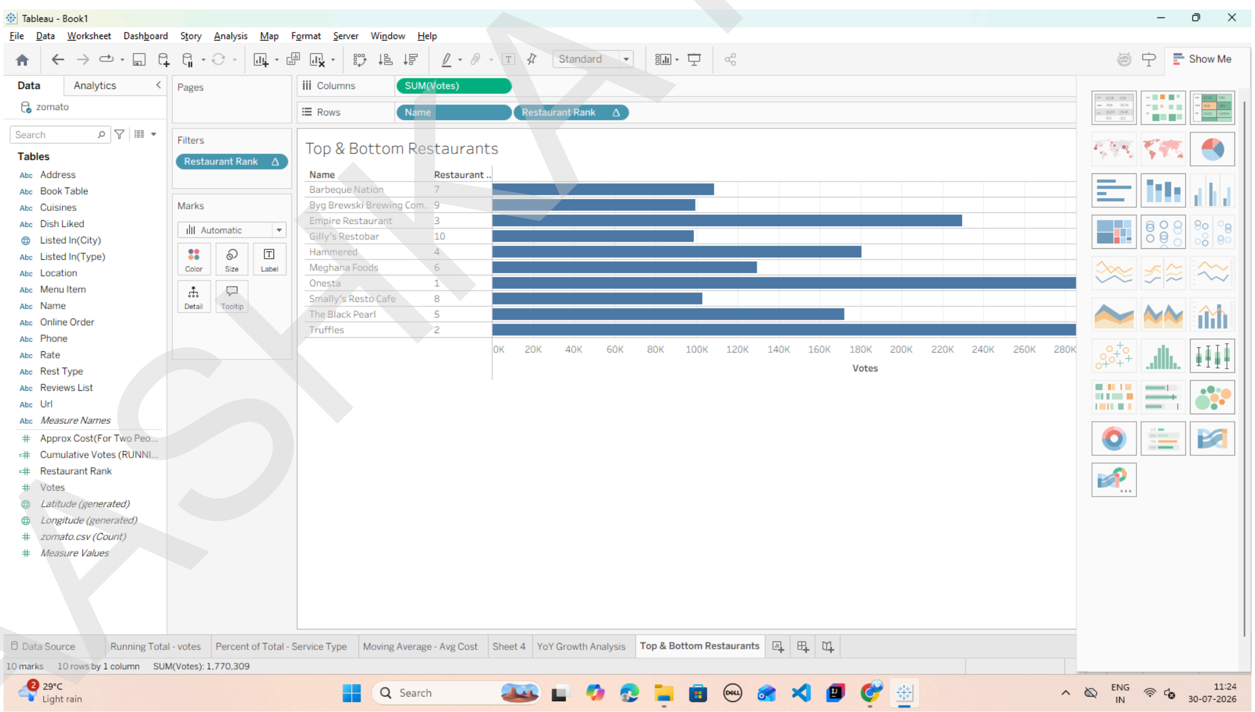
2. Convert Restaurant Rank to a Discrete field:



Task 3: Filter Dynamically to Display Top N Only



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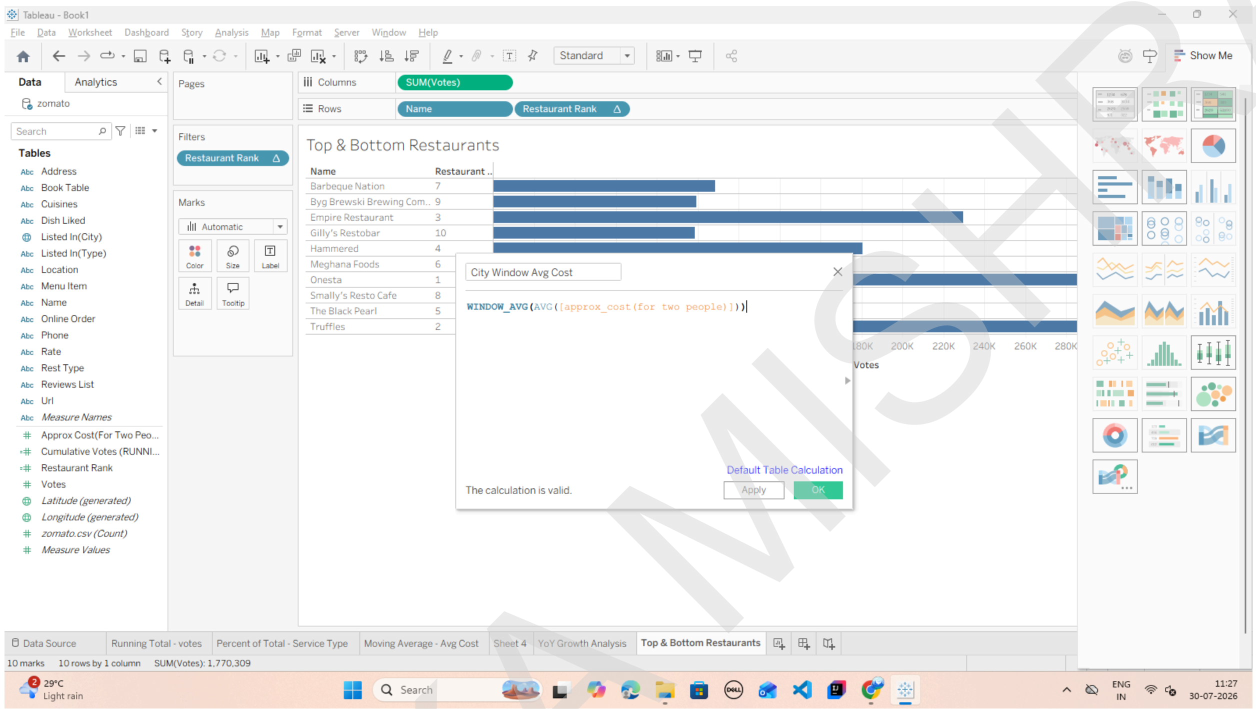
E: Implement window functions (WINDOW_SUM, WINDOW_AVG)

Task 1: Create a Window Average Calculation

1. Click the small drop-down arrow at the top right of the Data Pane (next to the Search bar) and select Create Calculated Field. 2. Name the field: City Window Avg Cost.
3. Input the following window function formula: WINDOW_AVG(AVG([approx_cost(for two people)]))



Task 2: Build the Window Comparison Visualization 1. Open a new worksheet and rename it: Cost vs City Average.
2. Drag listed_in(city) to the Rows shelf.
3. Drag approx_cost(for two people) to the Columns shelf. Change its aggregation to Average (Right-click pill $\rightarrow$ Measure (Sum) $\rightarrow$ Average).

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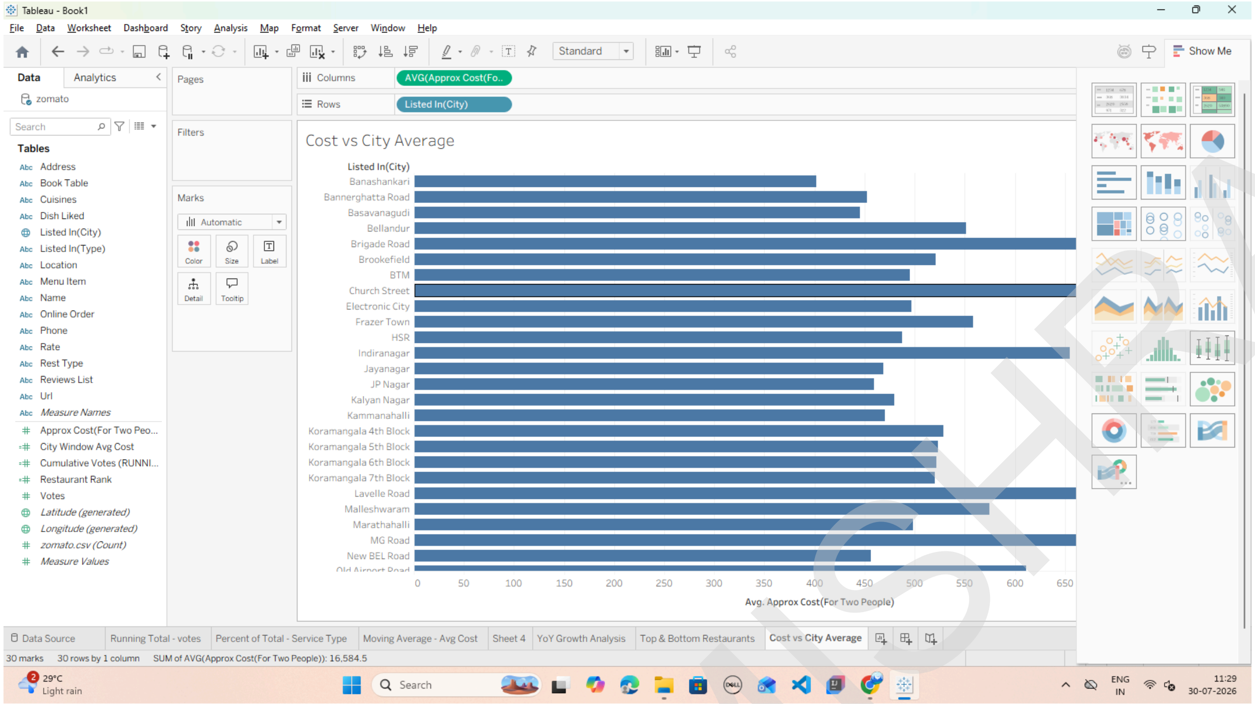
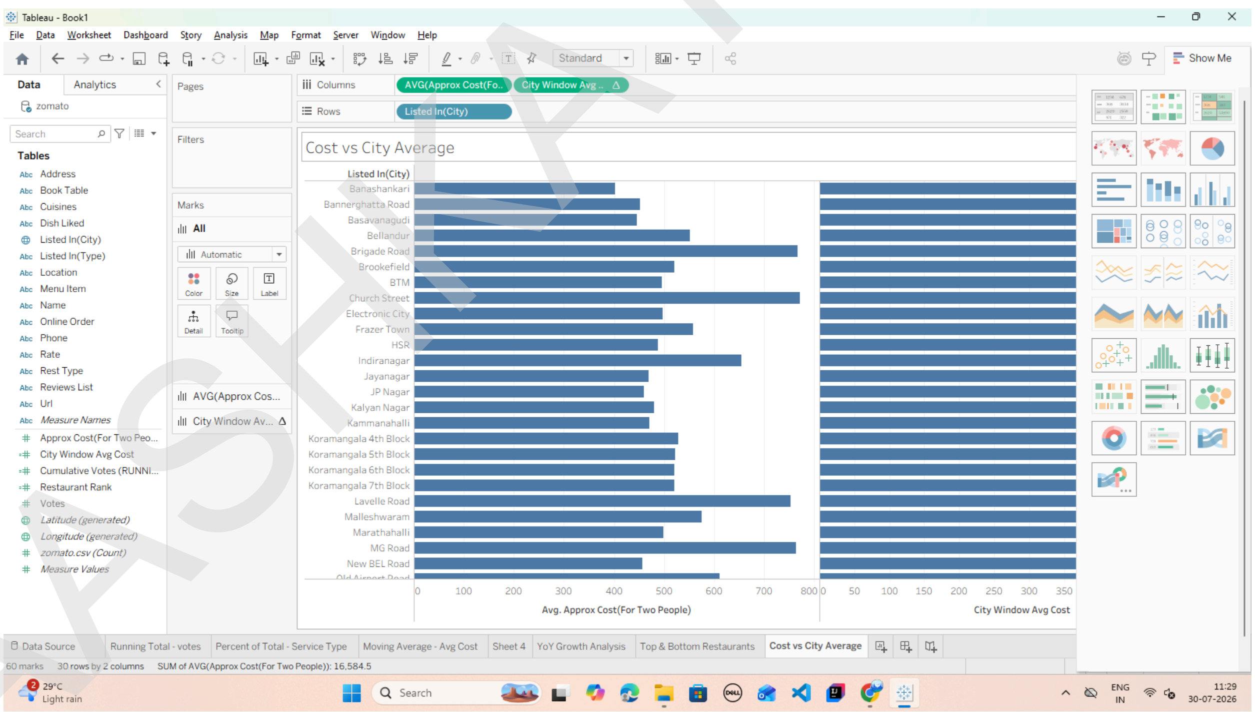


4. Drag your City Window Avg Cost calculated field onto the Columns shelf (placing it right beside AVG(approx_cost...)).



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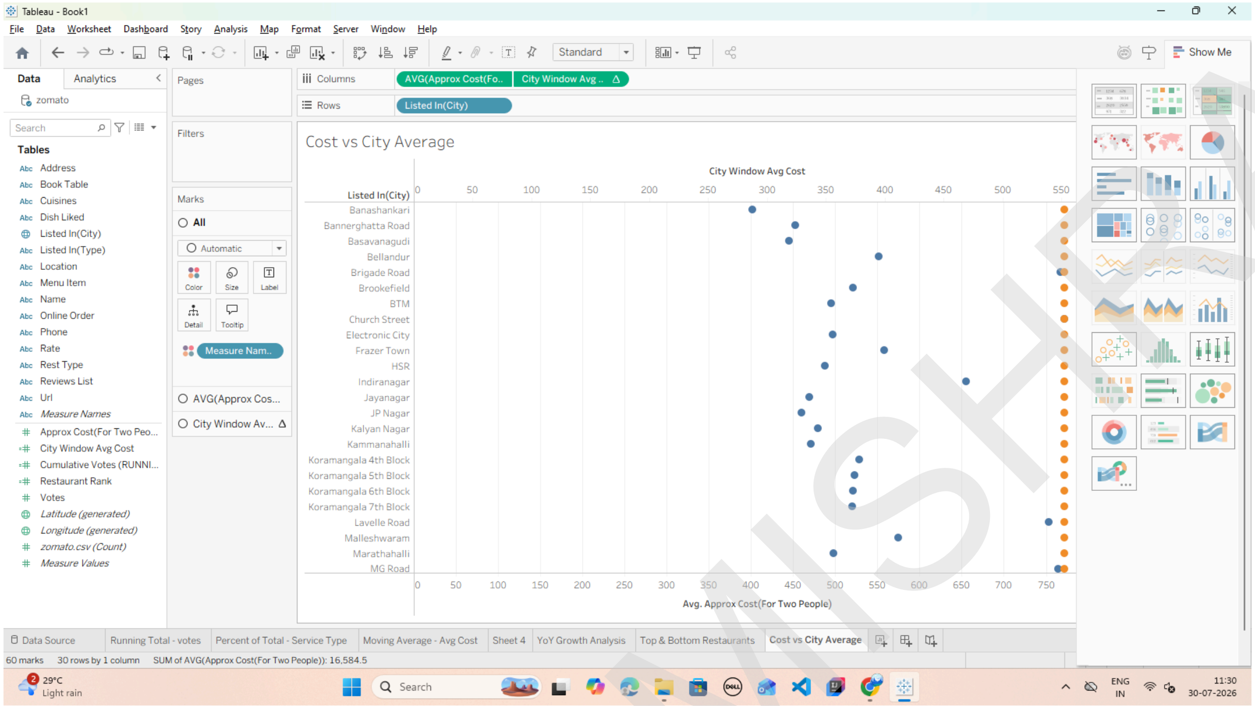
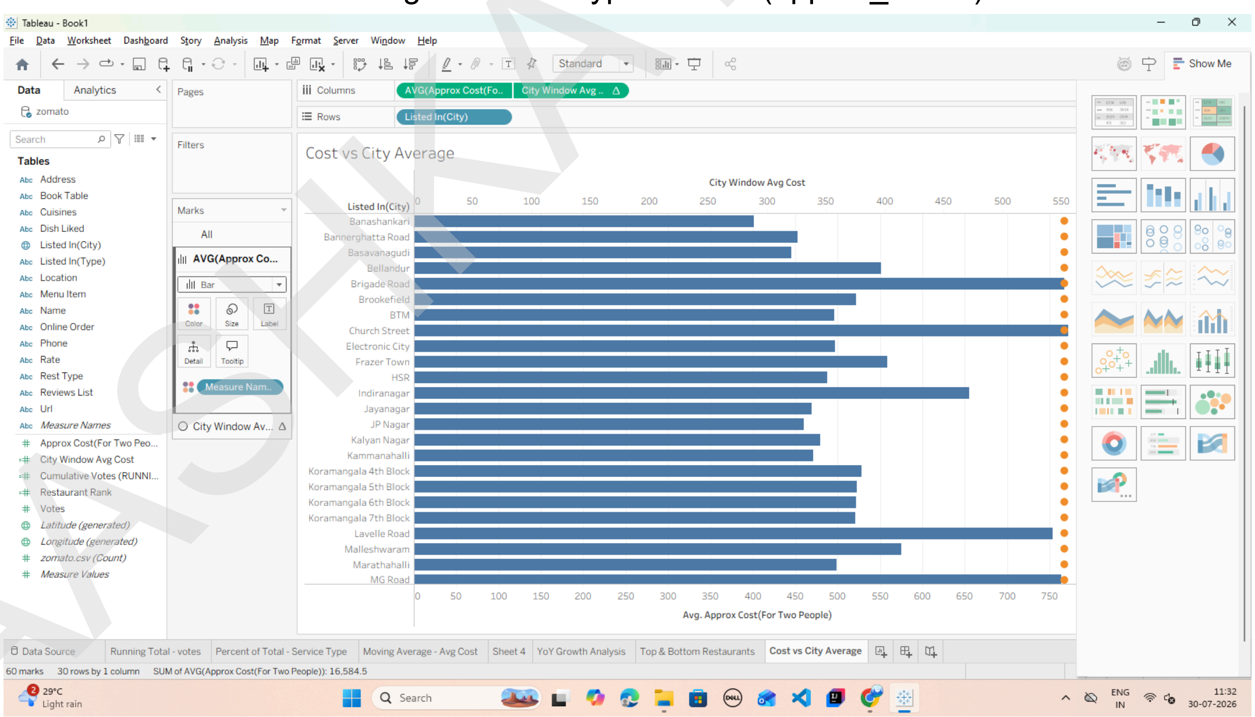
Task 3: Combine Measures into a Dual-Axis Reference View

1. Right-click the City Window Avg Cost pill on the Columns shelf and select Dual Axis.



3. In the Marks card: o Change the mark type for AVG(approx_cost...) to Bar.



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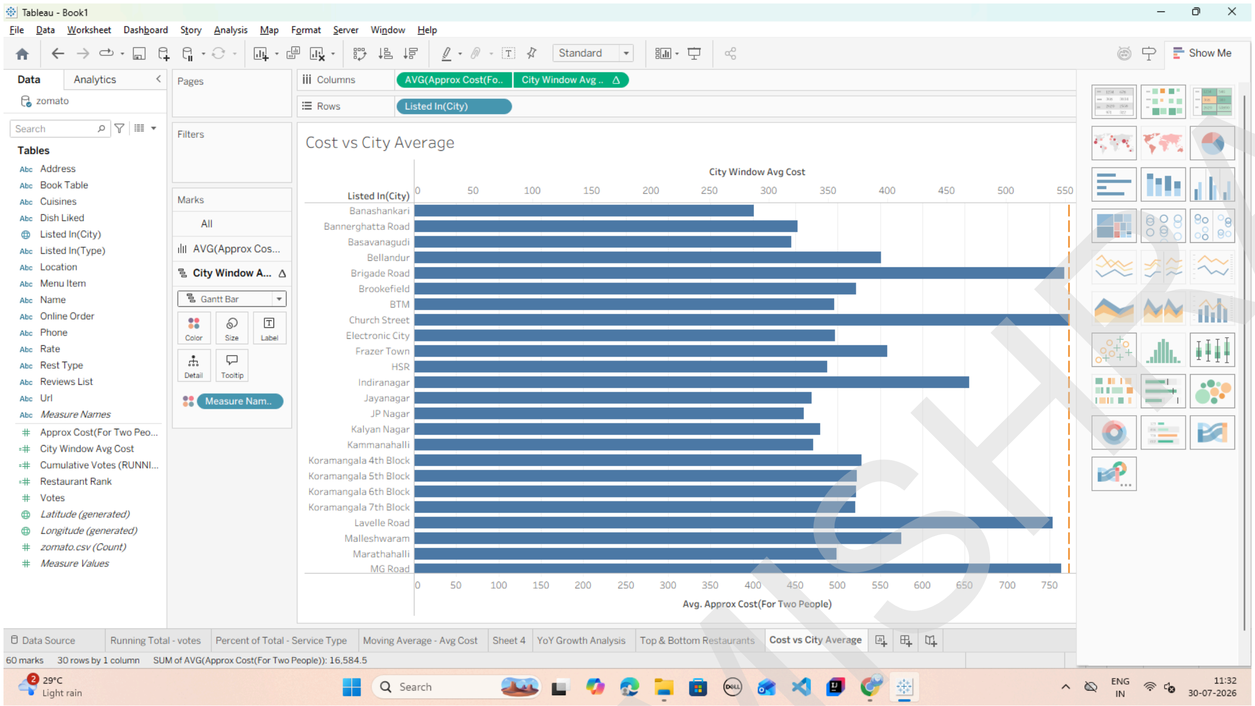
o Change the mark type for City Window Avg Cost to Line (or Gantt Bar)



Task 4: Create a Window Difference Metric (WINDOW_SUM)

1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field.
2. Name the field: Difference from Window Sum.
3. Input the following expression to compare each city's vote count against the total sum of all votes in the visible window: SUM([votes]) - WINDOW_AVG(SUM([votes]))

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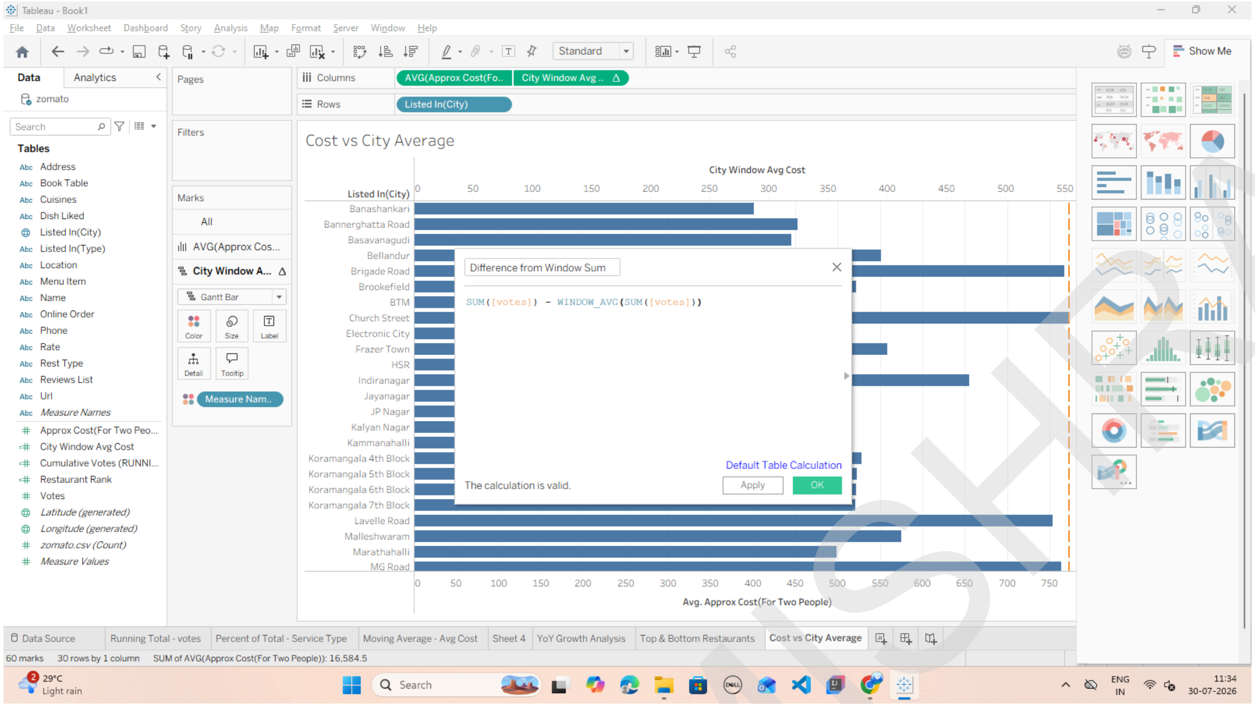
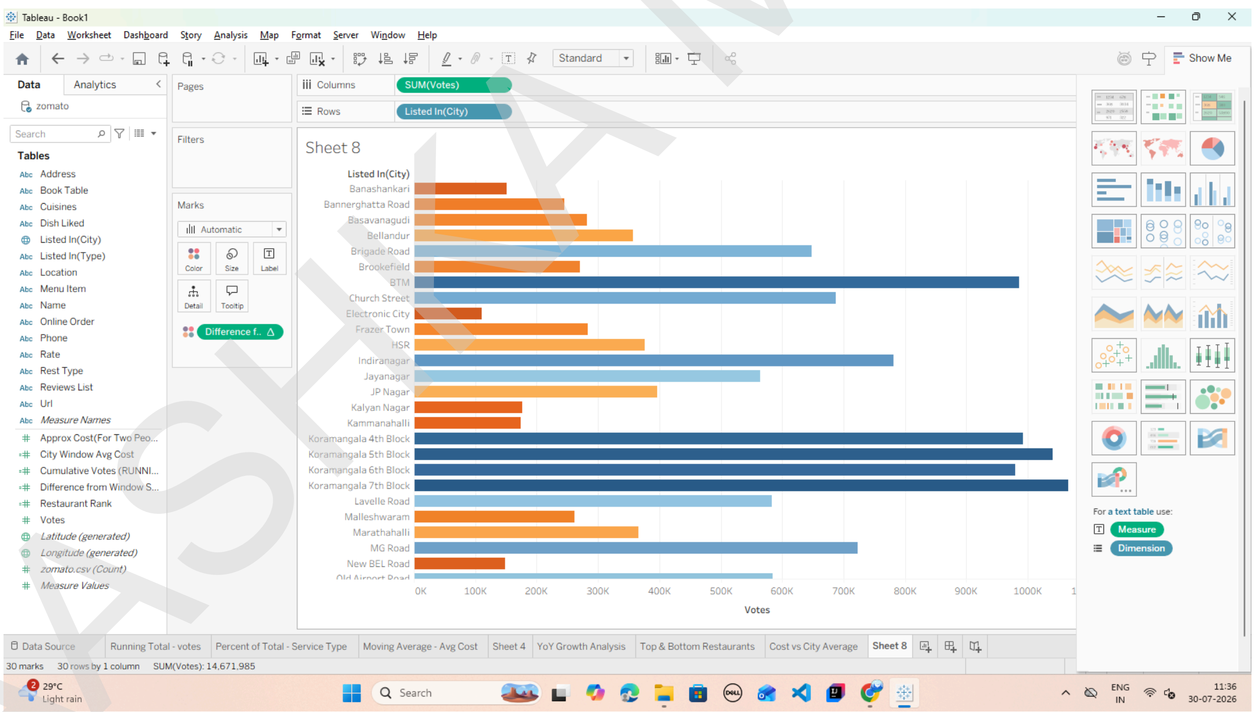


4. Apply it to the View:

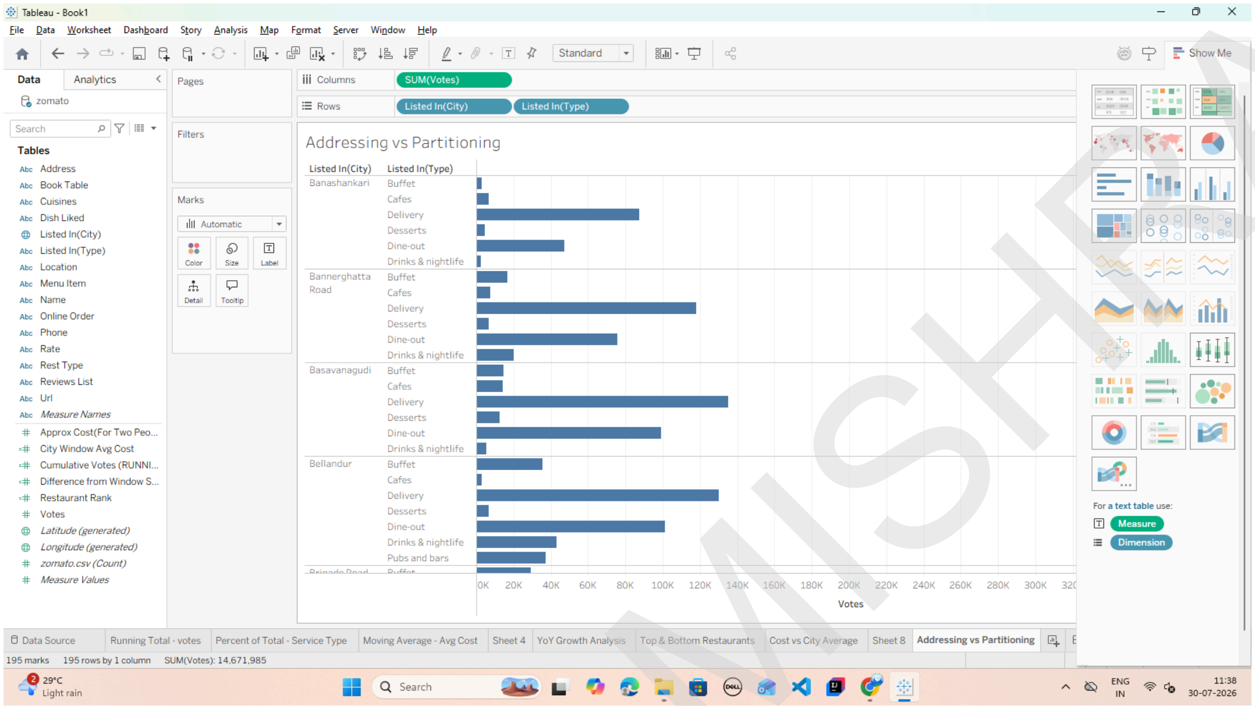


F: Customize addressing and partitioning

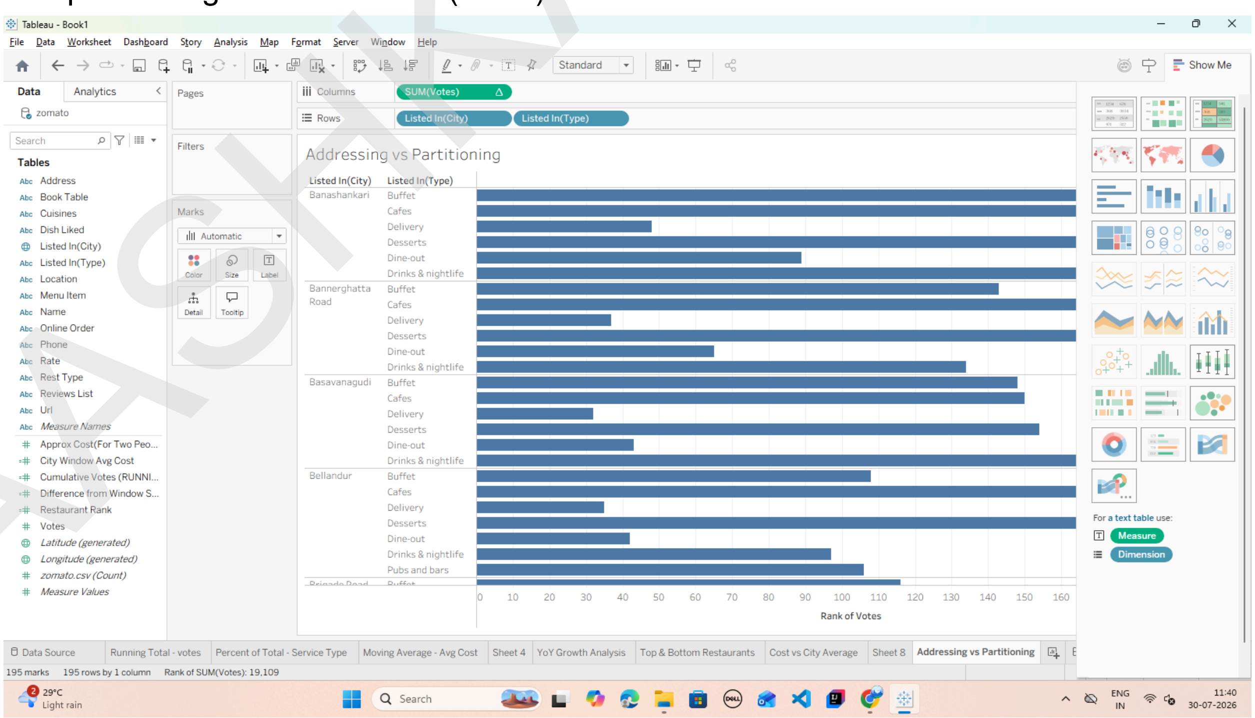
Task 1: Build a Nested View for City and Service Type

1. Open a new worksheet and rename it: Addressing vs Partitioning.
2. Drag listed_in(city) to the Rows shelf.

3. Drag listed_in(type) to the Rows shelf, placing it right next to listed_in(city) (creating a two-level nested hierarchy).
4. Drag votes to the Columns shelf as SUM(votes).



Task 2: Apply Partitioning via "Compute Using" Options
1. Right-click the SUM(votes) pill on the Columns shelf, hover over Quick Table Calculation, and select Rank. 2. Right-click the SUM(votes) pill again and hover over Compute Using: o Select Table (down):



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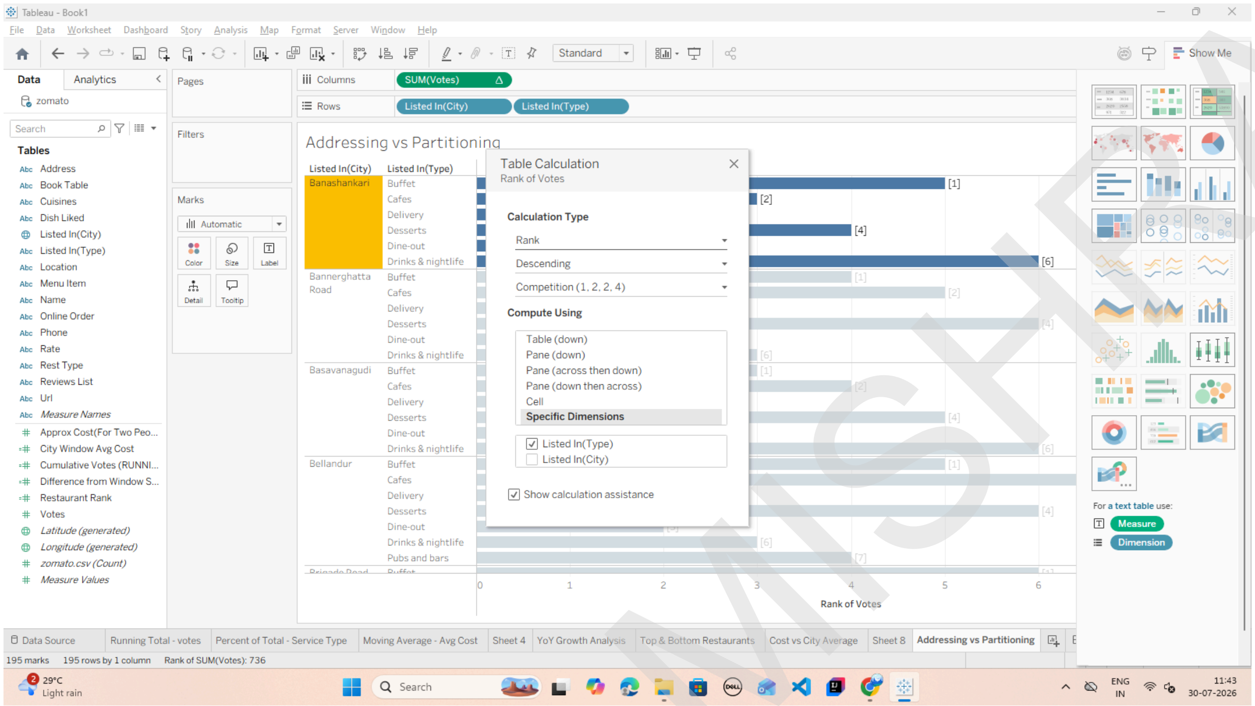
Task 3: Manually Define Addressing with Specific Dimensions

1. Right-click the SUM(votes) pill (with the Δ icon) on the Columns shelf and select Edit Table Calculation.



G: Create dynamic titles and labels using table calculations

Task 1: Create a Dynamic Label Parameter

1. Right-click anywhere in the Data Pane and select Create Parameter. or click on the small triangle drop down button next to search on the data pane and select Create Parameter

2. Name the parameter: Top N Threshold.

3. Set Data type to Integer.

4. Set Current value to 5. 5. Under Allowable values, select Range:

o Minimum: 3

o Maximum: 20

o Step size: 1

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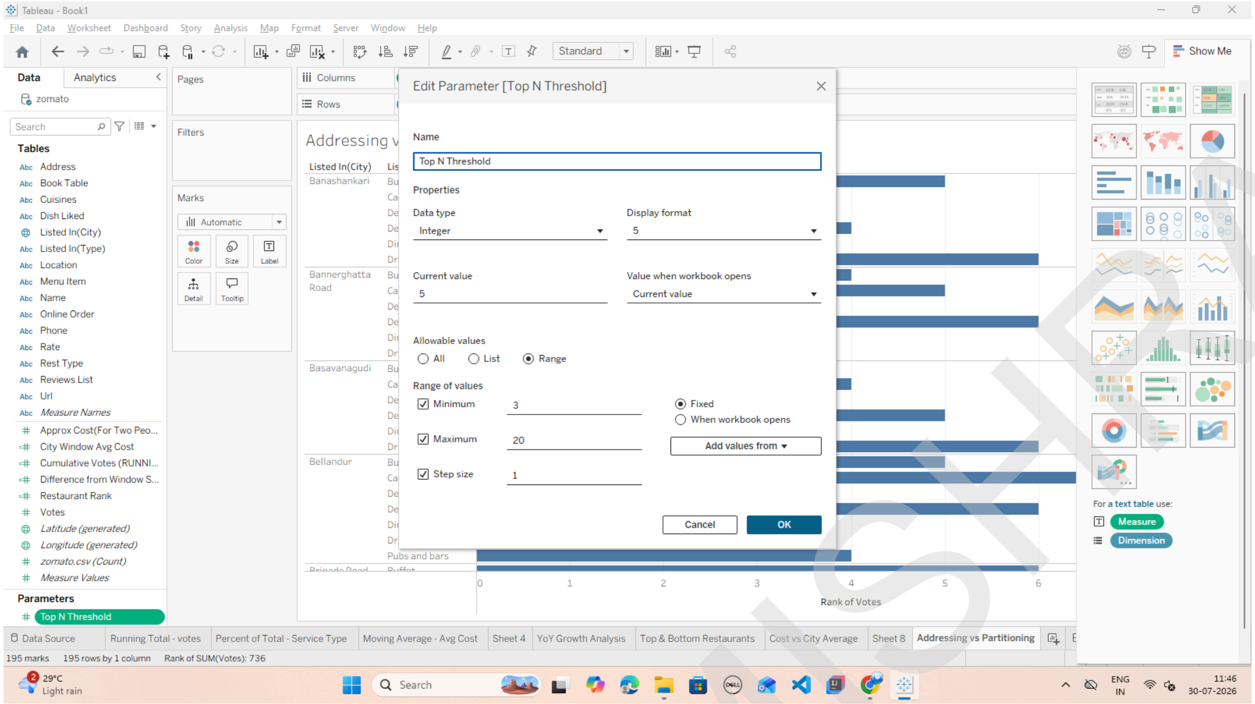
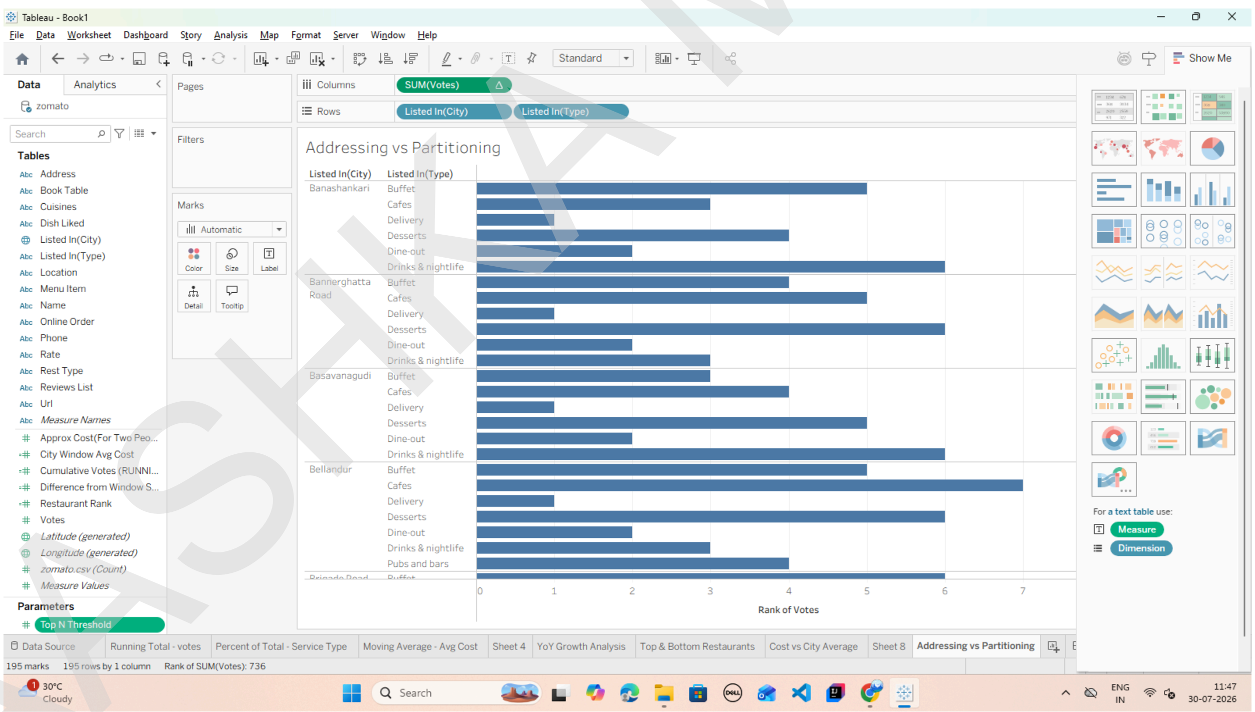


5. Right-click Top N Threshold in the Data Pane and select Show Parameter Control.



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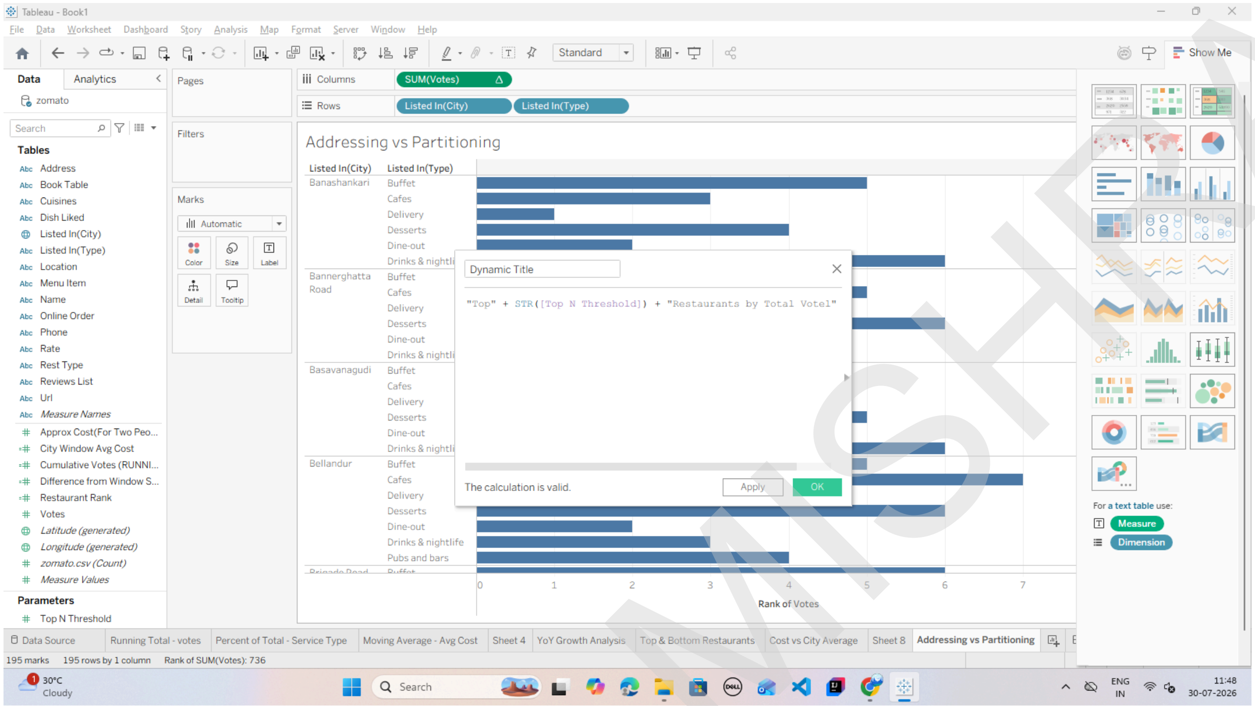
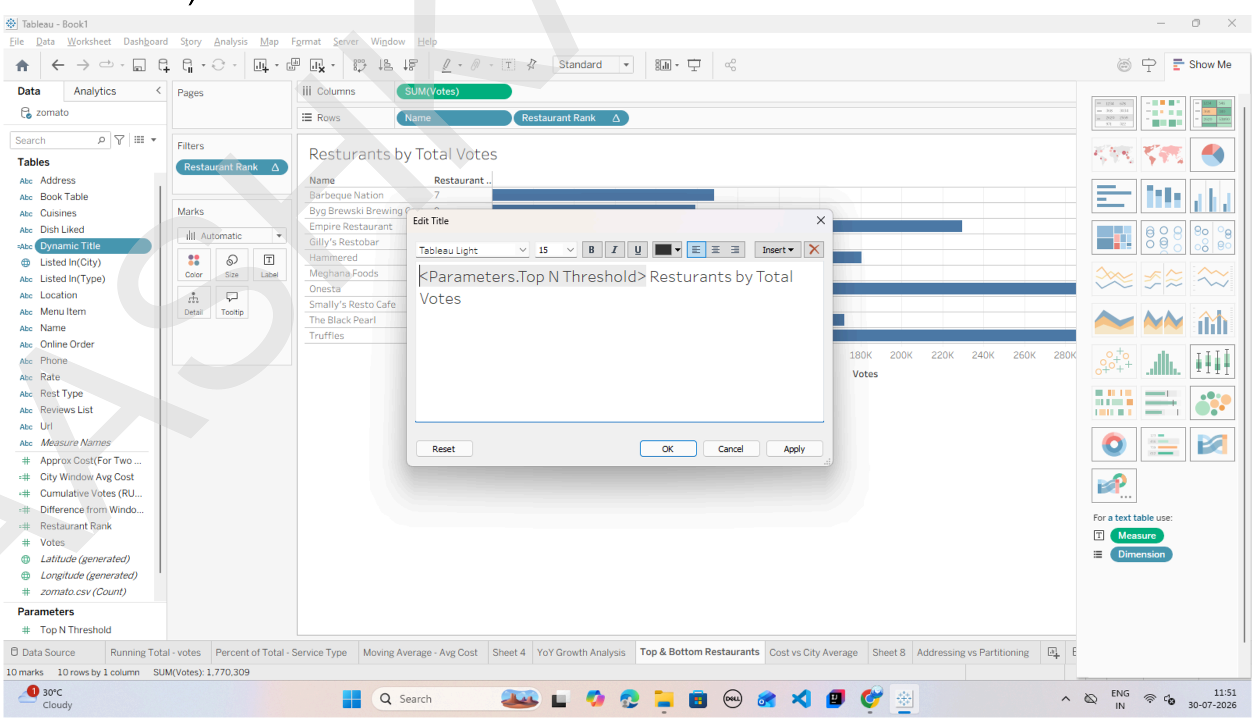
Task 2: Create a Dynamic Header Calculation

1. Create a new Calculated Field named Dynamic Title: "Top " + STR([Top N Threshold]) + " Restaurants by Total Votes"



Task 3: Embed Calculated Fields and Parameters into the Sheet Title

1. Double-click the Sheet Title at the top of your worksheet (e.g., Top & Bottom Restaurants).



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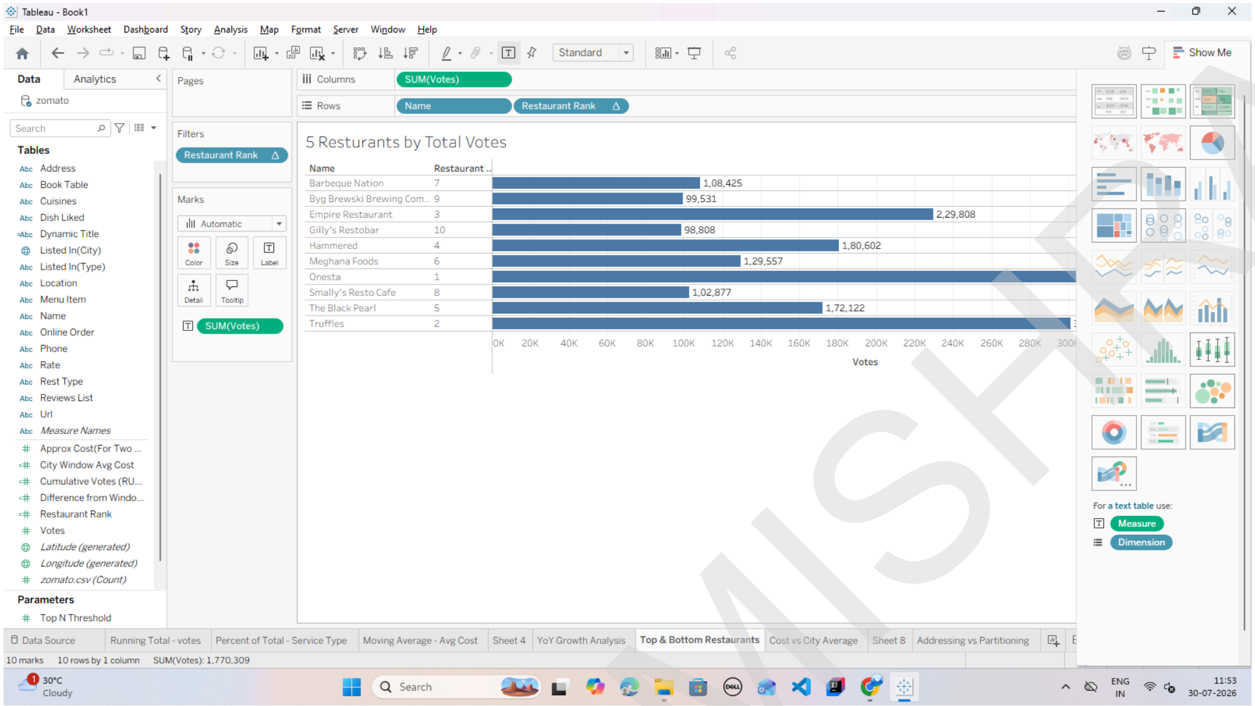
Task 4: Add Dynamic Mark Labels (Highlighting Extremes)

1. Drag SUM(votes) to the Label card on the Marks card.



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